IBM QUANTUM HUB · SCHOOL OF PHYSICS · THE UNIVERSITY OF MELBOURNE

Large Language Models

A physicist's perspective

Tobias J. Osborne · Leibniz Universität Hannover

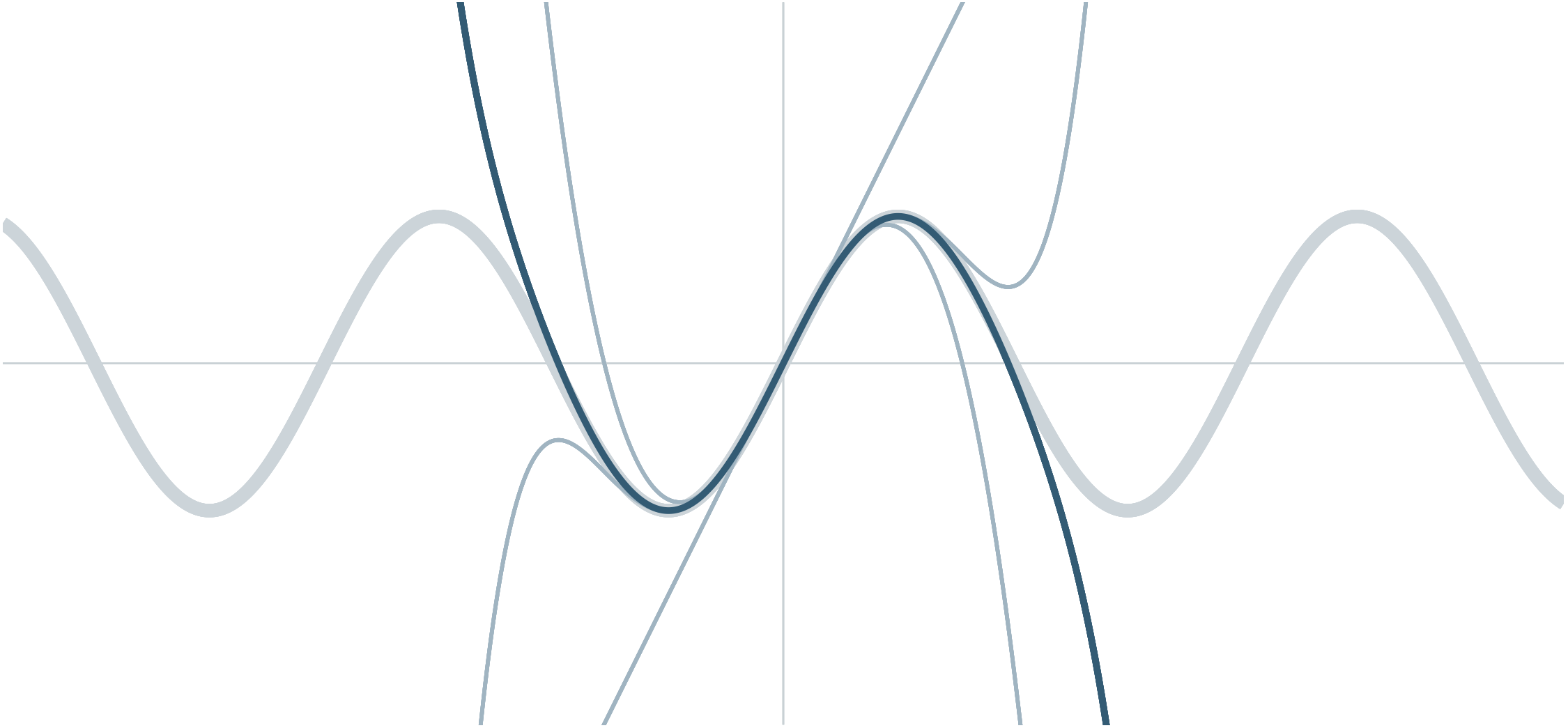
16 October 2026

$$\sin x \approx x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!}$$



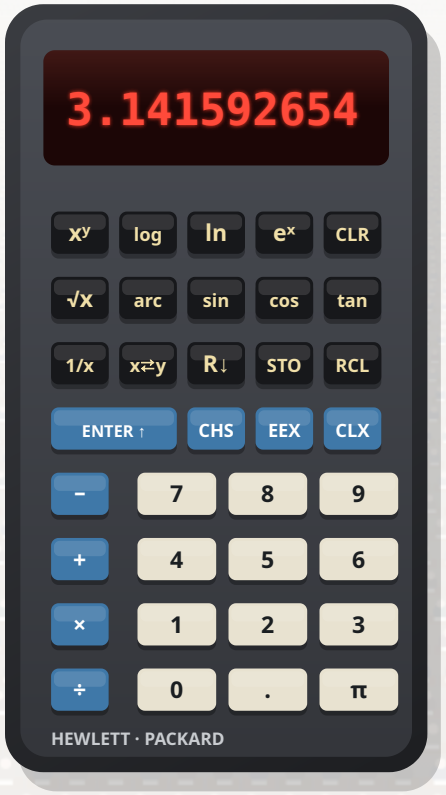
Colin Maclaurin
1698 to 1746

4
terms, by hand



64,000 computers

L. F. Richardson’s forecast factory, imagined in 1922



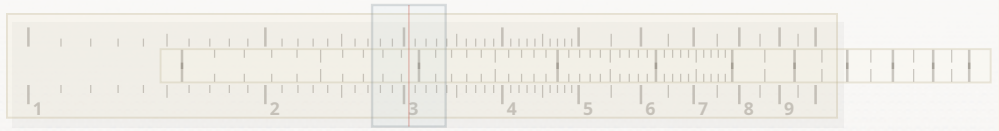
HP-35

January 1972

$395

100,000 sold in 1972

the sales goal was 10,000



1976 TI-30

under $25

1,700,000,000,000

times the performance of computing by hand, at least

Richardson (1922), quoted by P. Lynch (1993) · Computer History Museum · The Henry Ford · Nordhaus (2007), lower end of his range

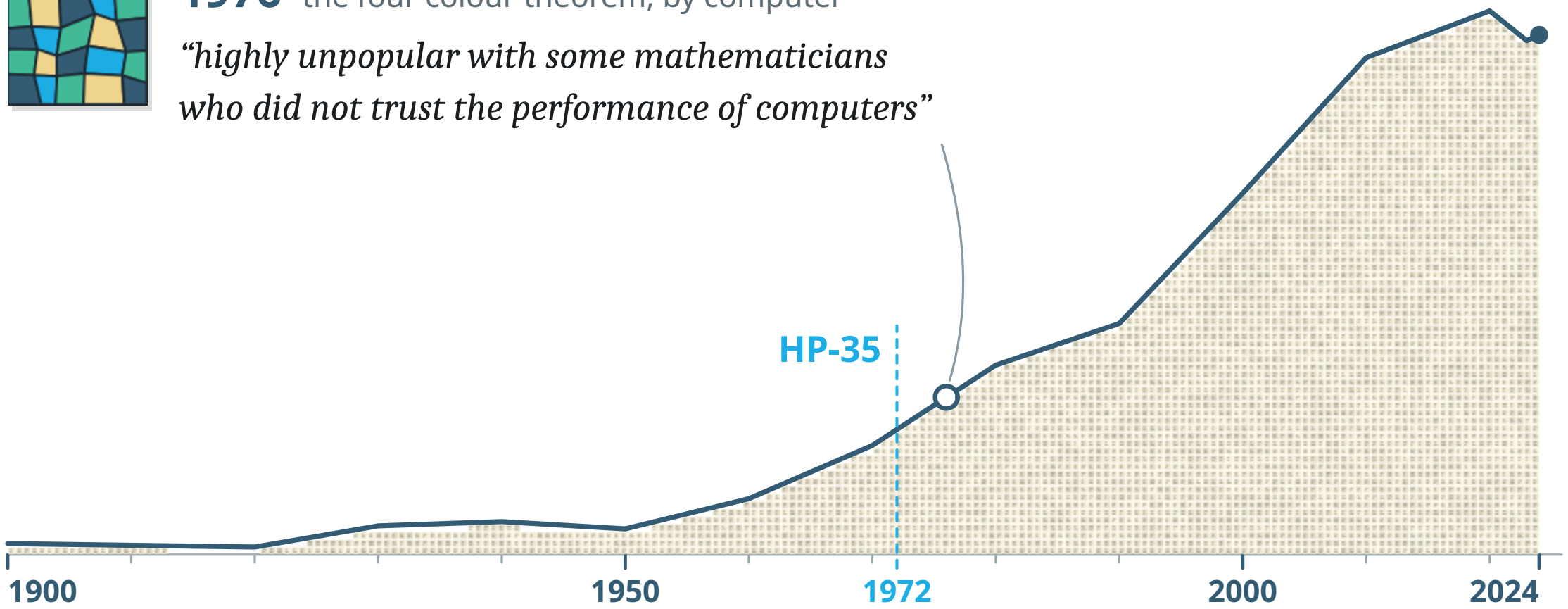
124,917

mathematical publications in the year 2024

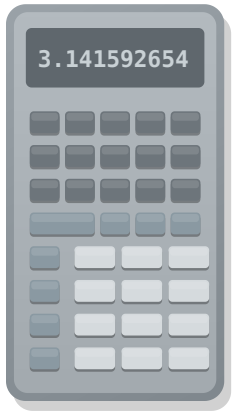


1976 the four colour theorem, by computer

“highly unpopular with some mathematicians who did not trust the performance of computers”



zbMATH Open, records by publication year, read 12 Oct 2026; before 1940 from the *Jahrbuch*, coverage uneven · quotation: *The Boston Globe*, 30 Apr 2013 · tiles are texture



HP-35
January 1972
$395

6 October 2026

719 manuscripts,
written by a machine



github.com/openai/math: first commit 6 Oct 2026; 722 manuscripts then, 719 after three withdrawals · HP-35: Computer History Museum, HP

Quick survey: hands up

- Who has used ChatGPT, Claude, or Gemini?
- Who has used one for research?
- Who has used a coding agent?

PART ONE

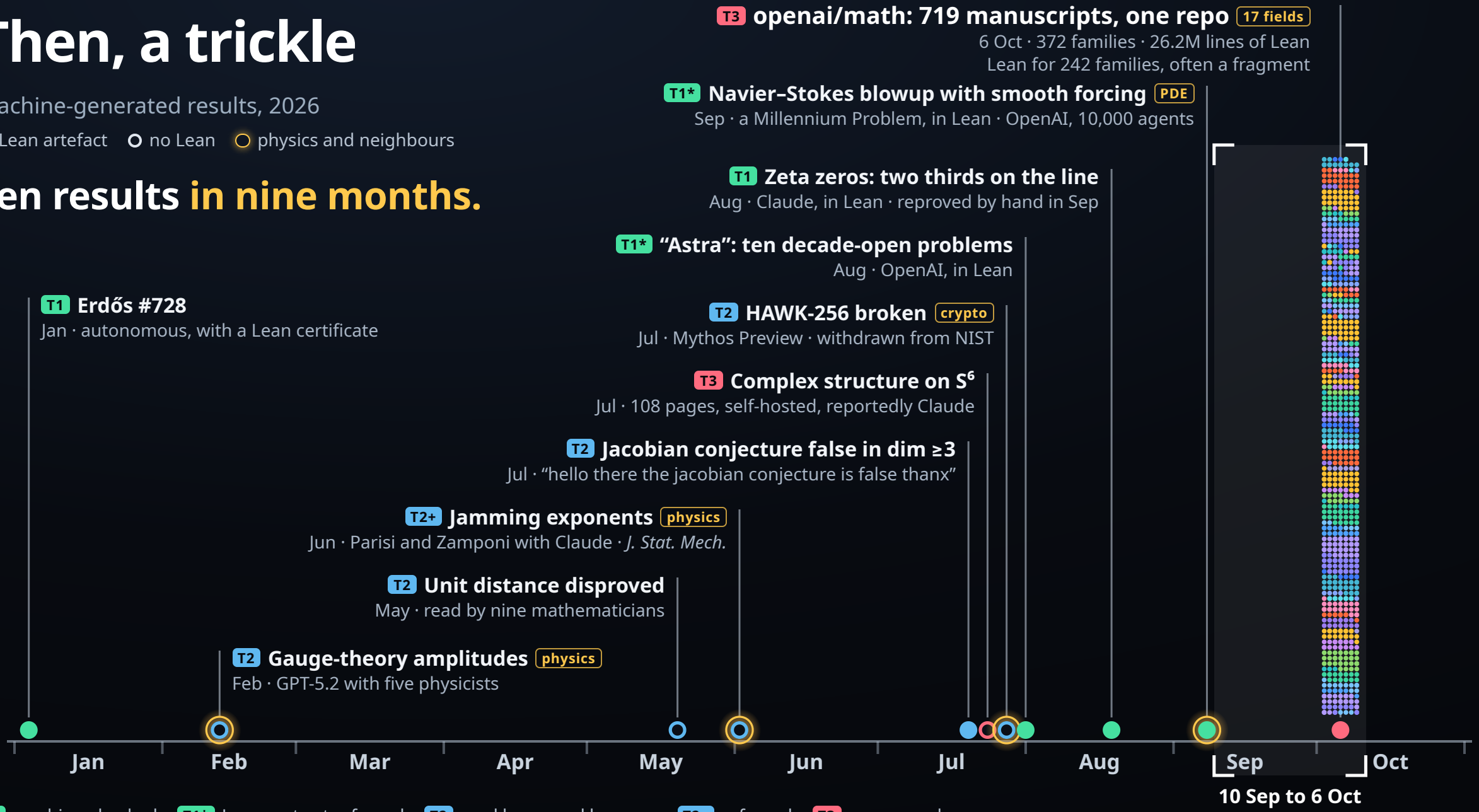
Why this talk?

Then, a trickle

machine-generated results, 2026

● Lean artefact ○ no Lean ○ physics and neighbours

Ten results in nine months.



T1 machine-checked T1* Lean, not yet refereed T2 read by named humans T2+ refereed T3 announced

dates of first announcement · arXiv, *J. Stat. Mech.*, lab and press announcements, github.com/openai/math (read 12 Oct 2026)

The trickle became a flood

github.com/openai/math · one dot per manuscript, by first folder date

719
manuscripts

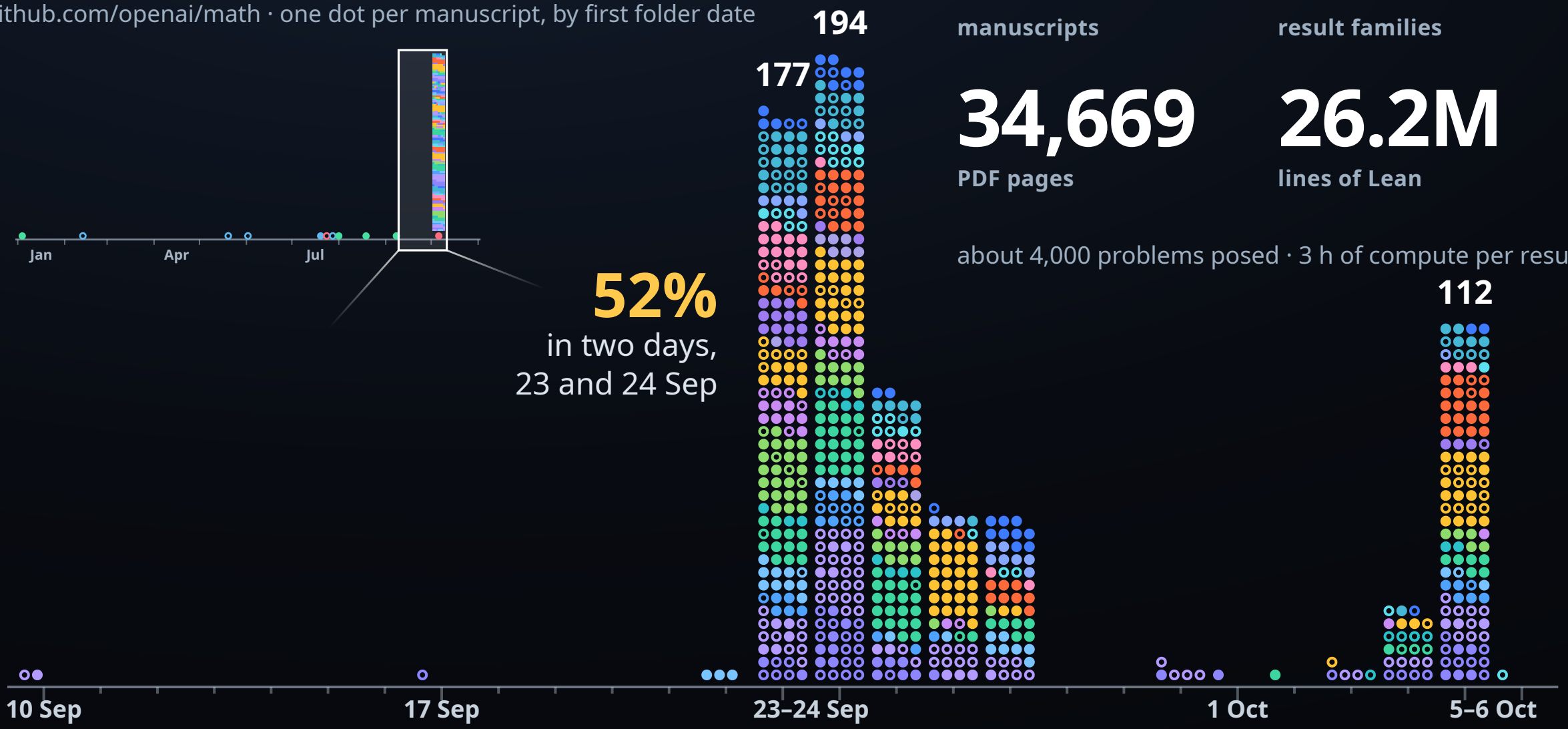
372
result families

34,669
PDF pages

26.2M
lines of Lean

about 4,000 problems posed · 3 h of compute per result

52%
in two days,
23 and 24 Sep

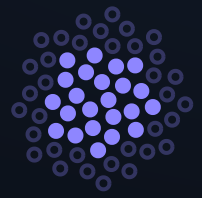


some Lean · no Lean · physics · everything else

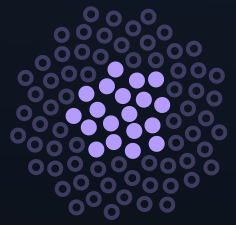
github.com/openai/math at commit fd4aeeb (8 Oct 2026), cloned 12 Oct 2026 · folder dates, first edition · totals: README and a file scan

17 disciplines

Some Lean for **65%** of families, often a fragment.
MUB(6) = 3: the Lean proves N(6) ≤ 5.



Number theory
55% Lean



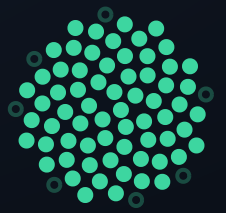
Algebraic & complex geometry
22% Lean



Real & complex analysis
63% Lean



Convex & metric geometry
87% Lean



Theoretical CS
85% Lean



Dynamics & ergodic
75% Lean



Combinatorics
89% Lean



Algebra
56% Lean



Math logic
100% Lean



Group theory
86% Lean



Topology
17% Lean



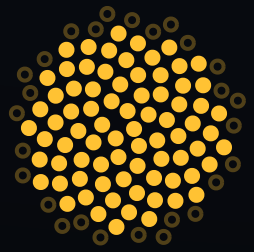
Functional analysis
91% Lean



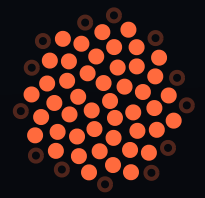
Differential geometry
52% Lean



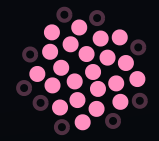
PDE
69% Lean



Probability & stat mech
66% Lean



Math physics
68% Lean

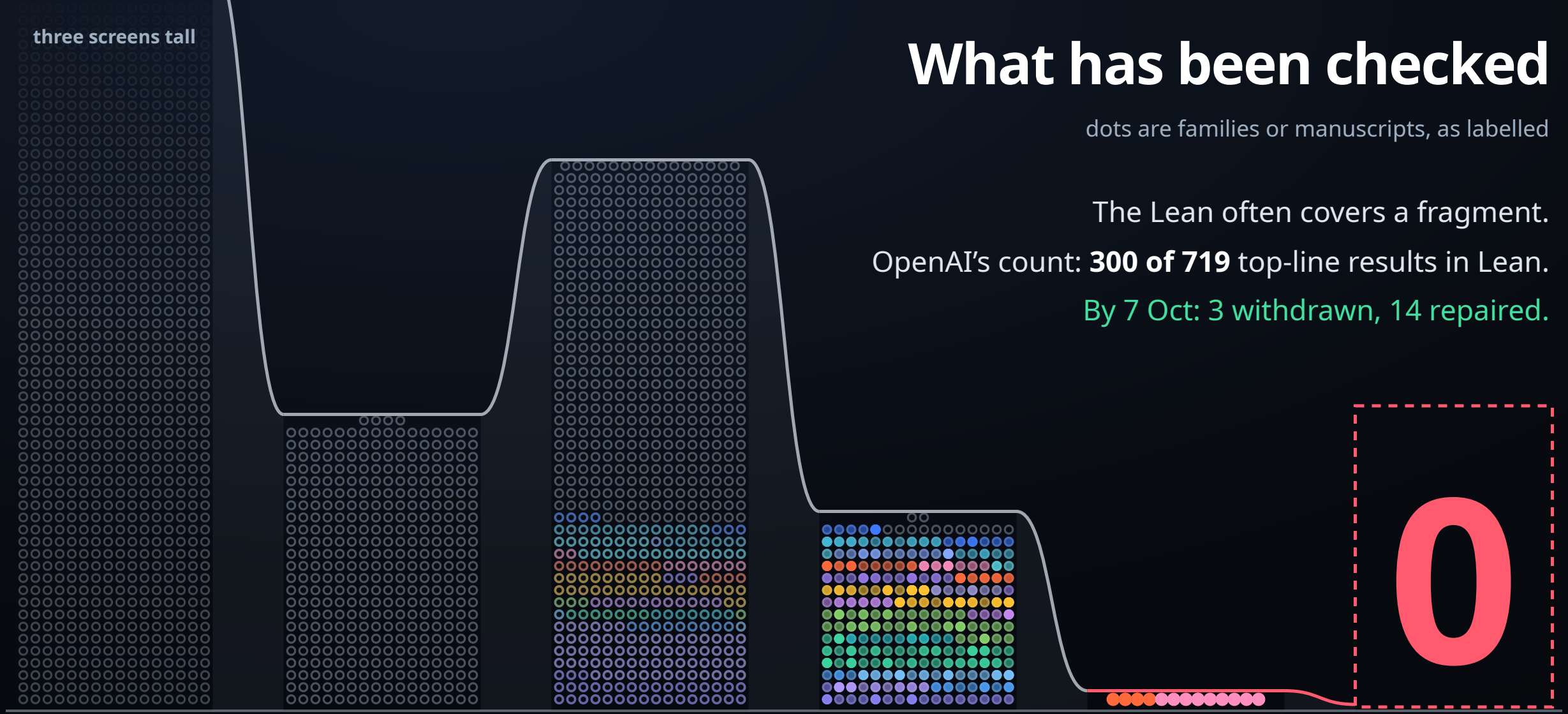


Operator algebras
79% Lean

github.com/openai/math at commit fd4aeeb (8 Oct 2026), cloned 12 Oct 2026 · disciplines as in overview.tex · Lean = an entry in lean/docs

What has been checked

dots are families or manuscripts, as labelled



The Lean often covers a fragment.

OpenAI's count: **300 of 719** top-line results in Lean.

By 7 Oct: 3 withdrawn, 14 repaired.

~4,000
problems posed

372
result families

719
manuscripts

242
families with some Lean

13 of 73
physics families with the headline in Lean

refereed
none found, 12 Oct

github.com/openai/math at commit fd4aeeb (8 Oct 2026), cloned 12 Oct 2026 · README, history.md, lean/docs · my count: physics blocks only

The physicist's picks

8 of 372 families
statements read, proofs not checked

FAMILY 265

2D area law

$S(A) \le C|\partial A|$ on $\mathbb{Z}^2$, from a gap alone.

◐ LEAN: §2 ONLY

FAMILY 268 COMPUTER-ASSISTED

Spin-1 Haldane gap

Uniform gap, even rings, $L \ge 60$.

○ NO LEAN

FAMILY 271

Heisenberg ferromagnet

$d \ge 3$: order at low T, Bloch's $T^{3/2}$ law.

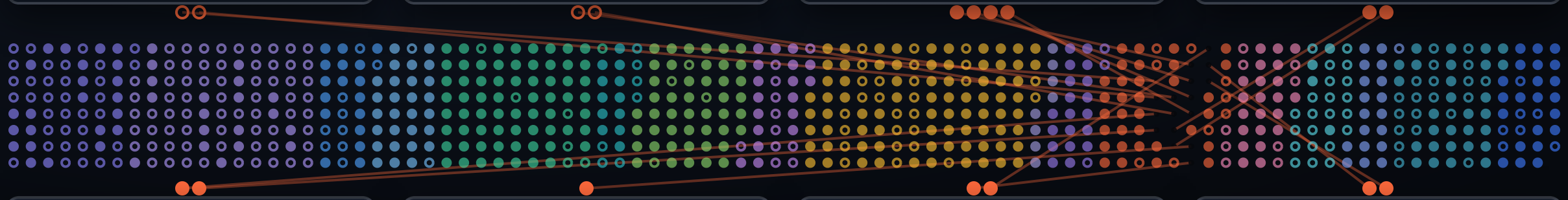
◐ LEAN: ORDER ONLY

FAMILY 269 COMPUTER-ASSISTED

Laughlin $\nu = 1/3$ gap

Uniform gap on the sphere, stable under disorder.

◐ LEAN: NO DISORDER



FAMILY 266 COMPUTER-ASSISTED

MUBs in dimension 6

N(6) = 3. Four bases excluded in binary64.

◐ LEAN: N(6) ≤ 5 ONLY

FAMILY 273

Entropy photon-number inequality

Guha, Erkmen, Shapiro: beam splitter.

● LEAN: HEADLINE

FAMILY 274

Parity $\notin$ QAC^0

Moore's conjecture, measured outputs.

● LEAN: HEADLINE

FAMILY 275

QMA-hard Coulomb energy

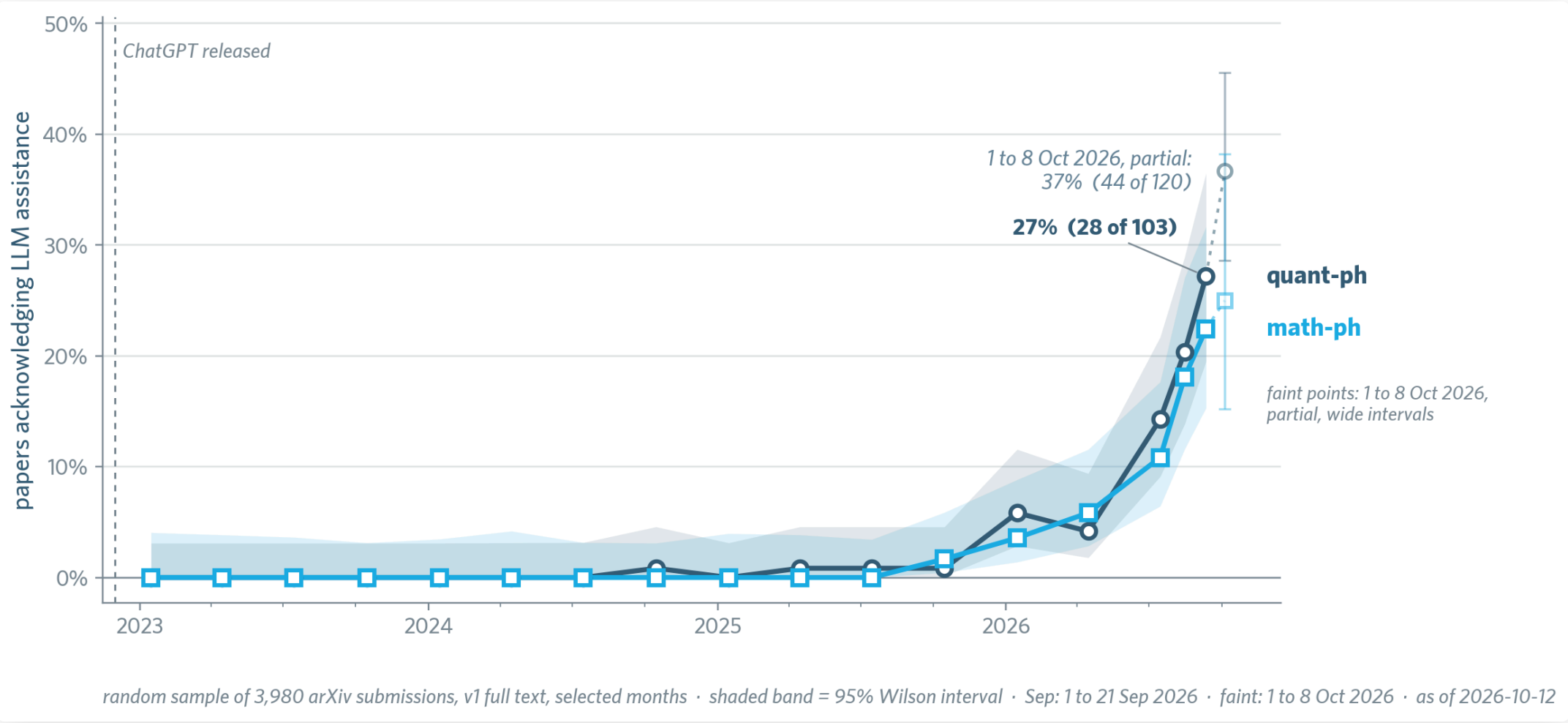
Continuum molecular Hamiltonian, unit charges.

● LEAN: BOTH THEOREMS

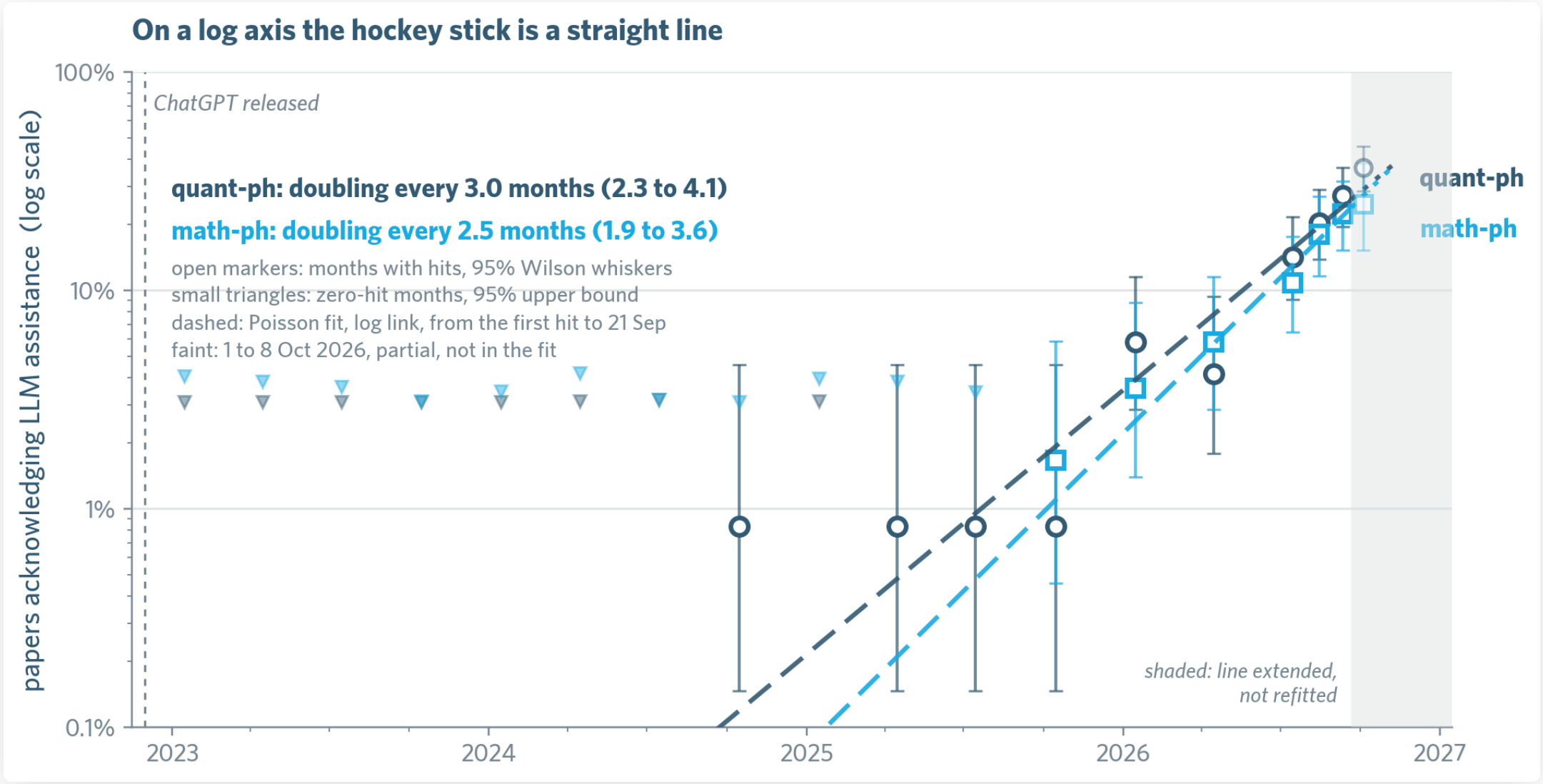
● HEADLINE ◐ FRAGMENT ○ NONE what the Lean covers

Headline in Lean: 3 of 8

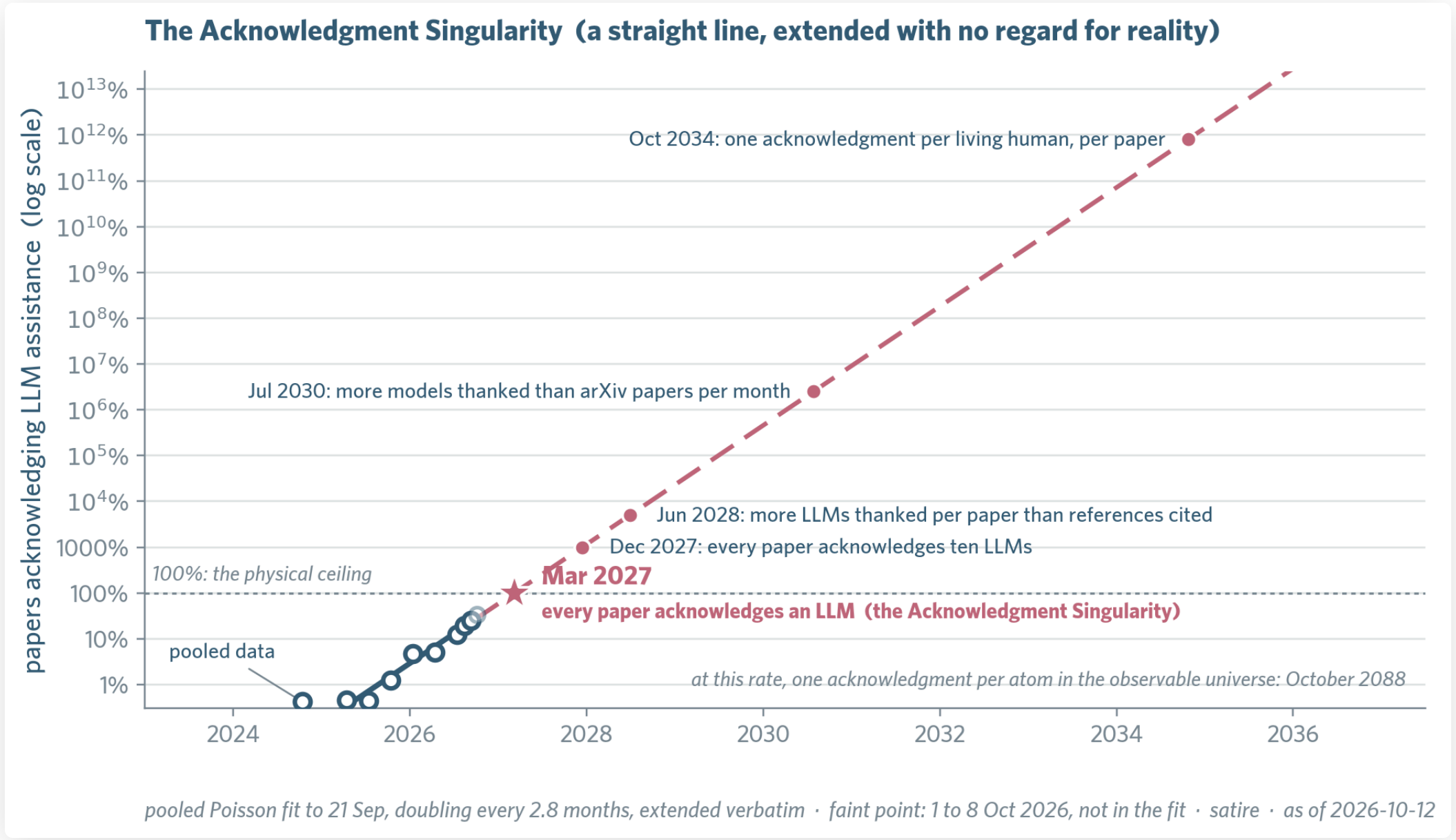
github.com/openai/math at commit fd4aeeb (8 Oct 2026), cloned 12 Oct 2026 · Lean status: lean/docs and my reading



Papers whose acknowledgments name an LLM: quant-ph **0% through 2023, 27% in September 2026**, 37% in 1 to 8 October
3,980 sampled v1 submissions, every hit read by hand · a disclosure rate, not a usage rate · faint: 1 to 8 Oct, partial · 12 Oct 2026



Same data, log axis: **doubling every 2.8 months** since the first hit (95% CI 2.3 to 3.5)
fit to 21 Sep, extended: 28% predicted for 1 to 8 Oct · observed 33% (95% CI 27 to 40) · consistent with the line, no bend yet · 12 Oct 2026


The Acknowledgment Singularity (a straight line, extended with no regard for reality)
papers acknowledging LLM assistance (log scale)
10^13%
10^12%
10^11%
10^10%
10^9%
10^8%
10^7%
10^6%
10^5%
10^4%
1000%
100%
10%
1%
Oct 2034: one acknowledgment per living human, per paper
Jul 2030: more models thanked than arXiv papers per month
Jun 2028: more LLMs thanked per paper than references cited
Dec 2027: every paper acknowledges ten LLMs
100%: the physical ceiling
Mar 2027
every paper acknowledges an LLM (the Acknowledgment Singularity)
pooled data
at this rate, one acknowledgment per atom in the observable universe: October 2088
2024
2026
2028
2030
2032
2034
2036
pooled Poisson fit to 21 Sep, doubling every 2.8 months, extended verbatim · faint point: 1 to 8 Oct 2026, not in the fit · satire · as of 2026-10-12

The lag collapsed: from three years to ten months

SOFTWARE ENGINEERING

Jun 2021
Copilot: a machine writes plausible code

Mar 2024
Devin hyped, then debunked

Feb 2025
Claude Code: the tooling loop closes

Q2 2025
Engineers: it is good at my own job

Nov 2025
Opus 4.5 first past 80% SWE-bench

37 months | 19 months | 11 months | 12 months | 10 months



MATHEMATICS & THEORETICAL PHYSICS

Jul 2024
IMO silver: AlphaProof

Oct 2025
Ten Erdős problems: claimed, retracted

Jan 2026
Erdős #728: model → Lean → human

May 2026
Unit distance falls; Gowers would take it for the Annals

Sep 2026
GPT-6 Astra ships; Navier-Stokes in Lean

6 Oct 2026
openai/math: 719 manuscripts

You are here

Melbourne, 16 October 2026

Déjà vu

	SOFTWARE ENGINEERING, 2026	MATHEMATICS, SEP TO OCT 2026
Understanding	“What would it take for you to be comfortable shipping code without reading and understanding it?” Gergely Orosz, LDX3 New York	“[…] there is no one to ask when we get stuck in a proof.” Bryna Kra, on openai/math, 8 Oct 2026
Review	“Everyone is playing the theater of doing reviews.” An engineer, quoted by Orosz	“[…] the current volume is drowning the reviewers.” Bryna Kra, 13 Sep 2026
Debt	“Everything is broken.” Orosz, on software reliability	“Math slop causes technical debt in mathematics […]” Henry Cohn, 15 Sep 2026

The capability is real.
Now let us understand the instrument.

PART TWO

What is an LLM?

A new kind of resource: general-purpose intelligence, bought by the token

(kind of)

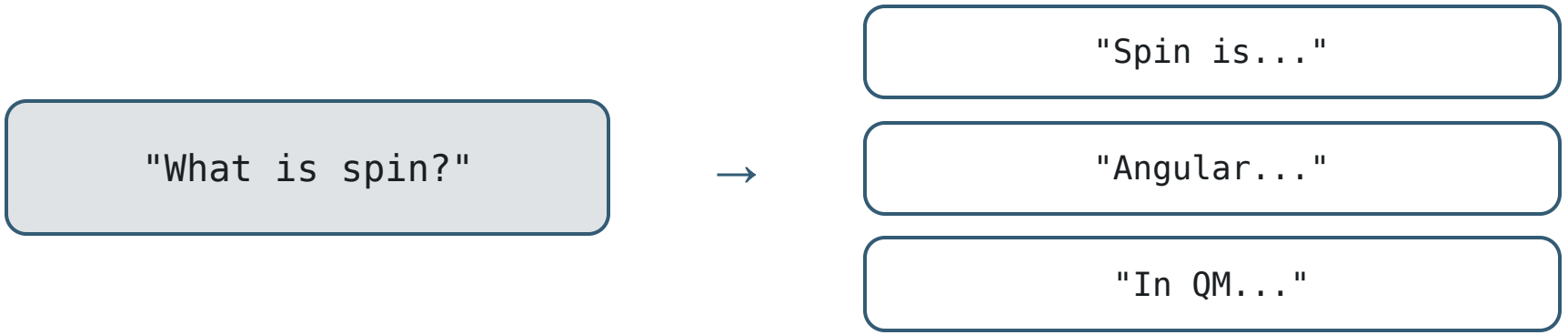
From outside, looking in

```
f : String -> String
```

Outside the API boundary, this is the observable behaviour.

No hidden state is carried between calls.

Nondeterministic



Same input, different outputs: the model samples from **P(next token | context)**

Temperature

$$P(\text{token}_i) = \exp(\ell_i/T) \,/\, \Sigma_j \exp(\ell_j/T)$$

spin	0%
charge	0%
mass	0%
momentum	0%
energy	0%
a	0%
color	0%
flavour	0%

Temperature 1.0

0.1 · ground state — 3.0 · hot

Sample **Autoplay**

> The electron has

Direct analogy to the Boltzmann distribution. Low T freezes into the ground state; high T melts toward uniform.

Tokens

Not characters. Not words.

Subword chunks from a fixed vocabulary **V** of about 100k entries.

```
'theoretical physics'  ->  ['theor', 'etical', ' physics']
```

More precisely: **f : Token* → Token***

An LLM is a stateless, nondeterministic function

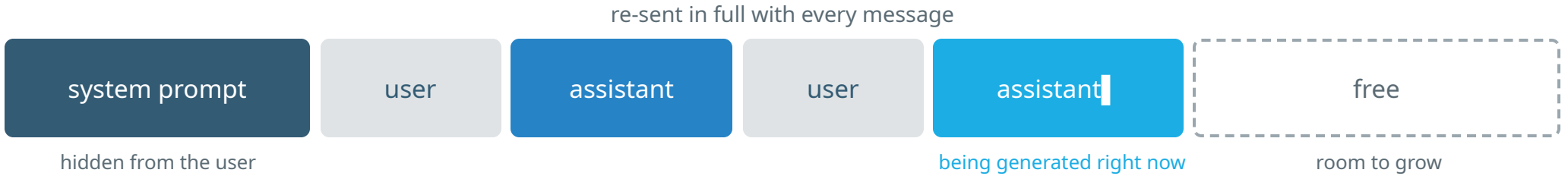
```
f : String -> String
```

that samples from P(next_token | context)

via a single forward pass through a frozen neural network

The context window

Everything the model sees is **one token sequence**, capped at a fixed length: the **context window**



Current models: **200k to 1M tokens** (a million tokens is roughly two thousand pages)

PART THREE

The illusion of chat

How chat works

```
Turn 1:  [user_1]
Turn 2:  [user_1, asst_1, user_2]
Turn 3:  [user_1, asst_1, user_2, asst_2, user_3]
```

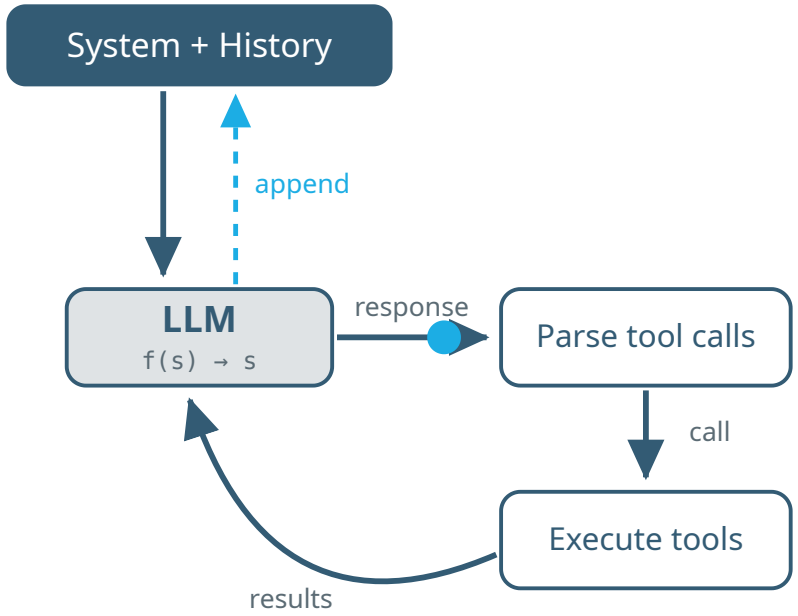
The client re-sends the **entire** history every turn: the LLM has no memory.

PART FOUR

From function to agent

The agent loop

```
while not done:
    response = llm(system_prompt + history)
    tool_calls = parse(response)
    results = execute(tool_calls)
    history += [response, results]
```



The complete architecture of every coding agent

The LLM never executes anything

It generates text; your scaffolding acts on it.

Closing the loop

Chat

closed loop, through *you*

you cut, paste, re-prompt

same loop, no human relay

→

Agent

closed loop, through tools

results feed back automatically

The agent closes the same loop faster, with no human cutting and pasting

Known failures, known fixes

~~finite context · no continual learning~~

Files as memory

✓

~~hallucination~~

Ground truth on disk

✓

~~nonzero error rate~~

Decompose and verify

✓

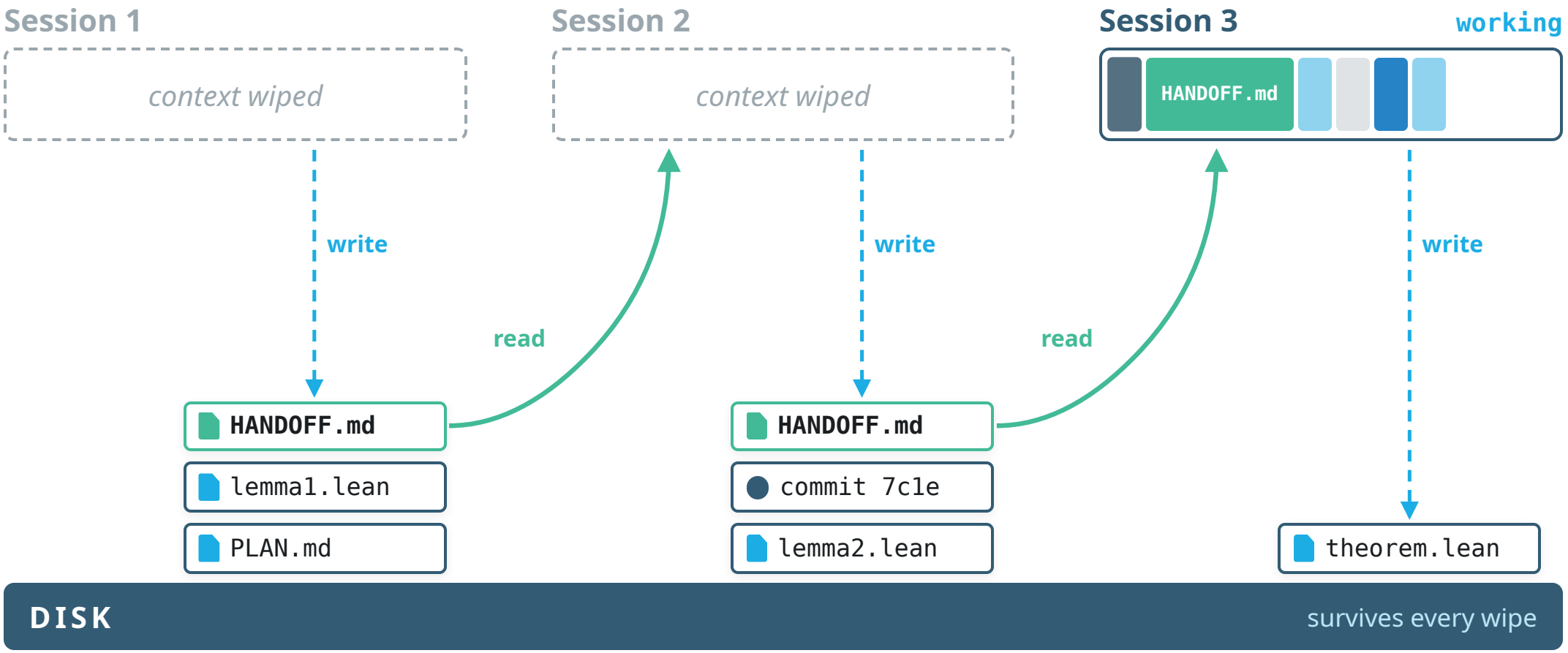
~~hyperfixation · sycophancy~~

Fresh adversarial check

✓

Reliable computation out of unreliable components

The filesystem is permanent memory



Every session forgets. The files do not.

Errors compound

A flat chain of thought has **no error detection**. If each step succeeds with probability 1 − ε:

$$P(\text{the whole derivation}) = (1 - \varepsilon)^n$$

PER-STEP ERROR	STEPS	SURVIVE INTACT
1%	100	37%

A slip at step 3 propagates silently to step 40, while the trace stays locally fluent throughout.

This is why long unchecked derivations rot

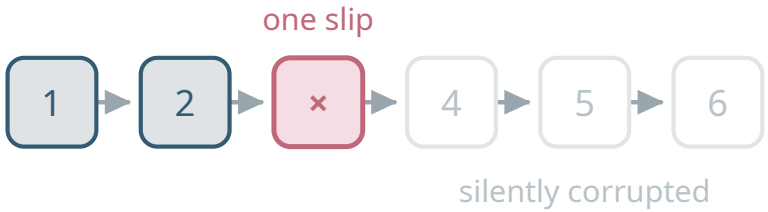
Subagents



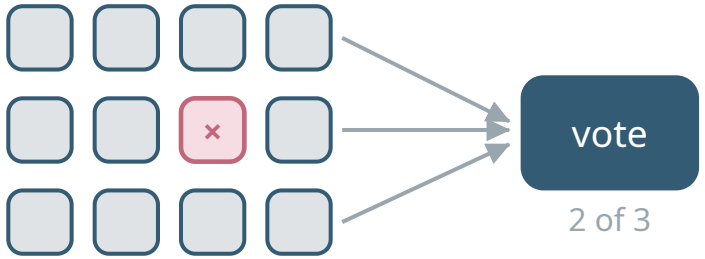
Each worker gets a clean window; results return as short reports.

Context isolation is the point

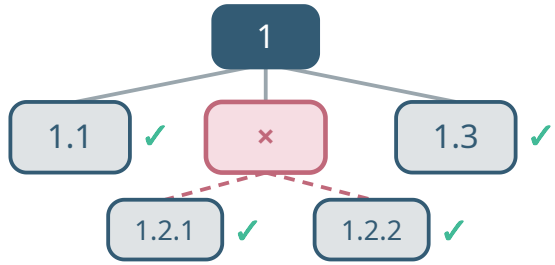
Three ways to spend compute on reliability



(a) flat chain. One slip at step 3; everything after it is quietly wrong.



(b) parallel repetition. Majority vote over final answers. Helps, but scales poorly with depth.



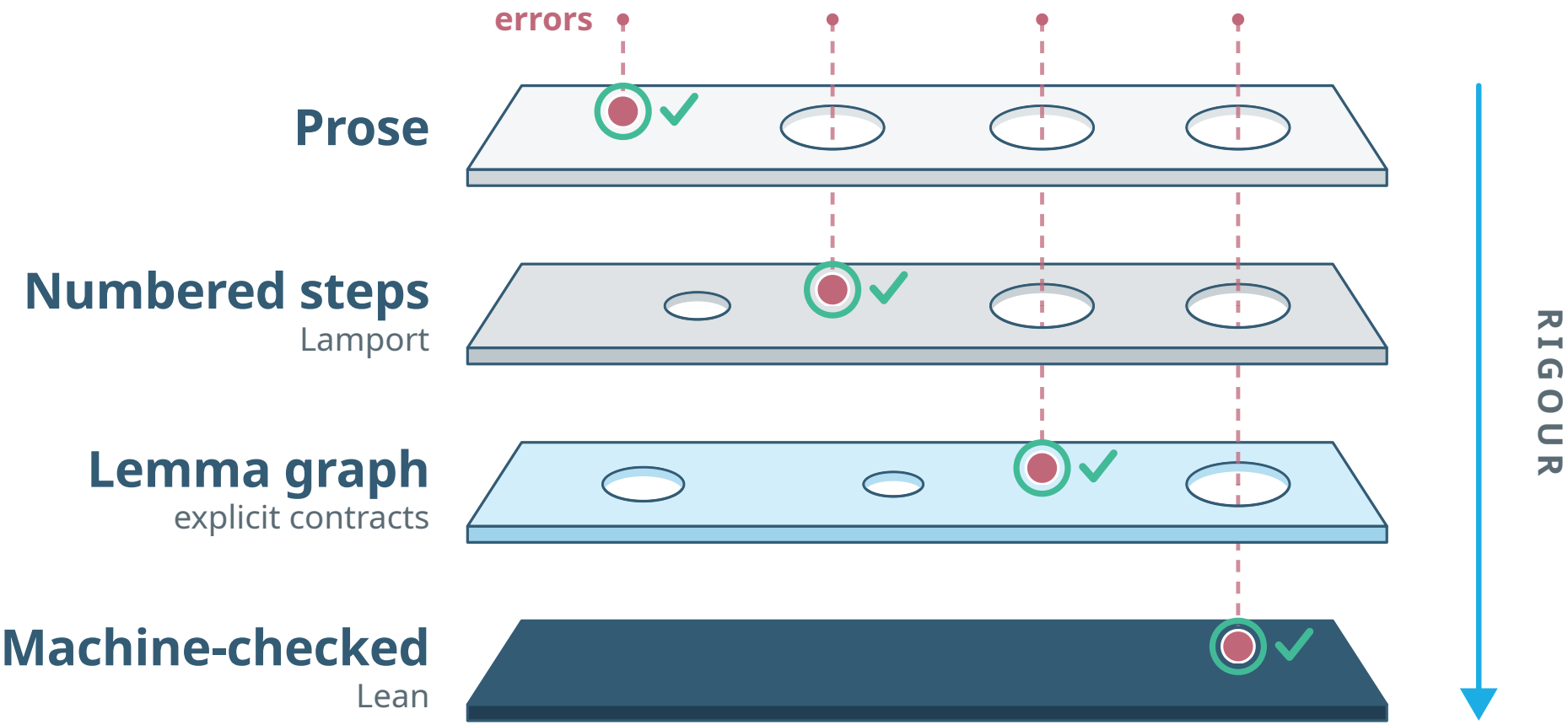
(c) recursive decomposition. Verify each block in isolation; expand the one that fails.

(c) is the concatenated-code strategy (von Neumann, 1956).

Above the verification threshold, reliability *compounds* instead of decaying

Defence in depth

Swiss cheese model (Reason, 2000)



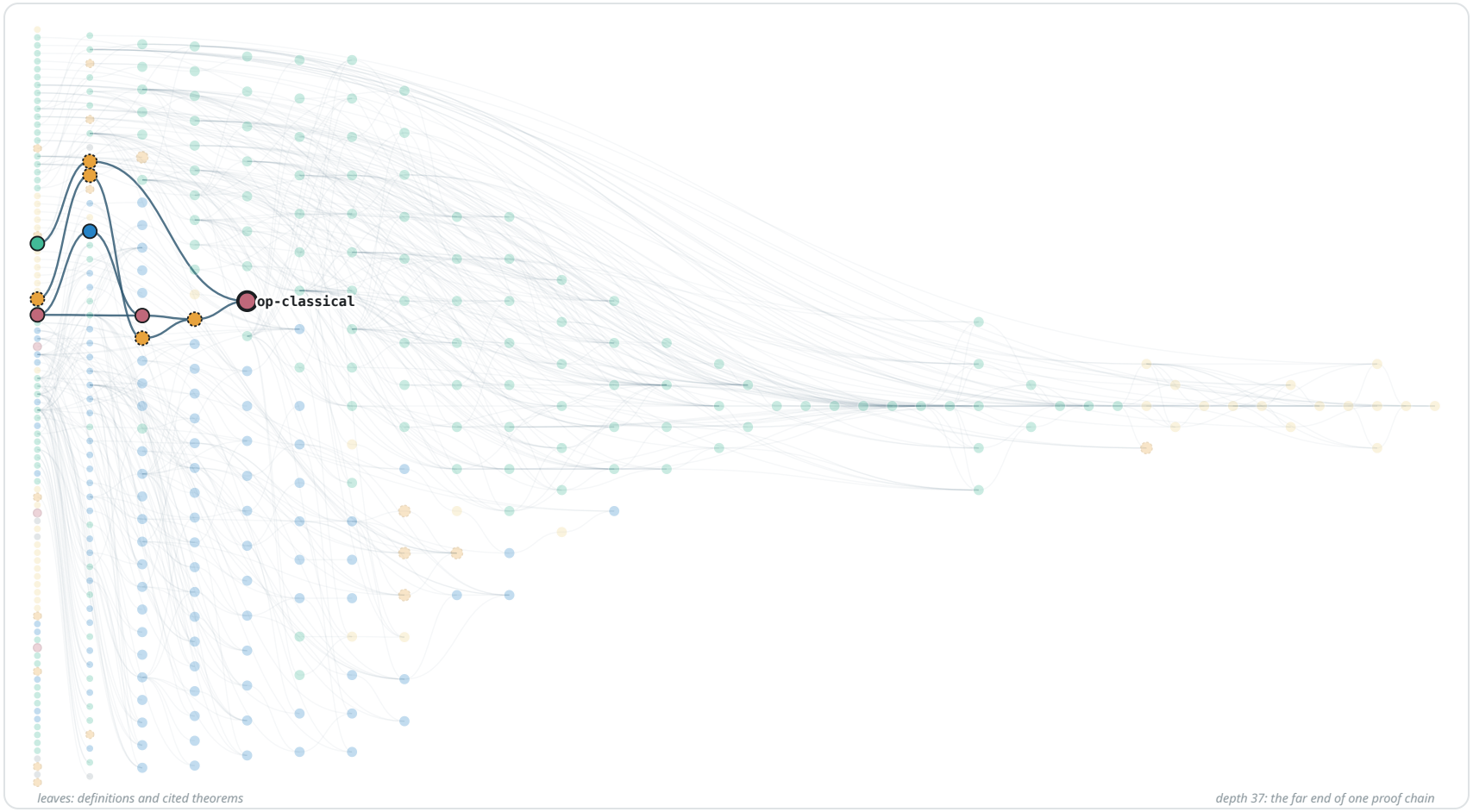
Prover → **Verifier** → **Formaliser** **adversarial agents push each claim one layer deeper**

schematic

github.com/tobiasosborne/vibefeld

What the Swiss cheese produces

one open problem, worked by agents for five weeks
364 results · 769 dependency edges · every arrow is a checked contract



click any node: what it rests on lights up · depth grows to the right

machine-validated: proved by one fresh agent, attacked by another	169
proved in prose, not yet adversarially checked	113
inherited paper-proof, unaudited here	22
conjecture, stated, heuristic or numerical	48
open problem	6
disproved or obstruction	6

op-classical

OPEN-PROBLEM OPEN

(OPEN) Classical projection stability: there are universal eta_0,C>0 (n-free) such that every row-stochastic Q with ||Q^2-Q||_{inf->inf} <= eta <= eta_0 admits a stochast...

rests on **9** results: **1** machine-validated · **1** prose · **7** unverified or open

The open problem itself. What it still rests on is orange and red: the gap is visible, named, and exactly where the next agent gets sent.

One agent proves, a fresh one attacks, a linker keeps the ledger. The colour is what survived.

Agents are amazing at helping you understand

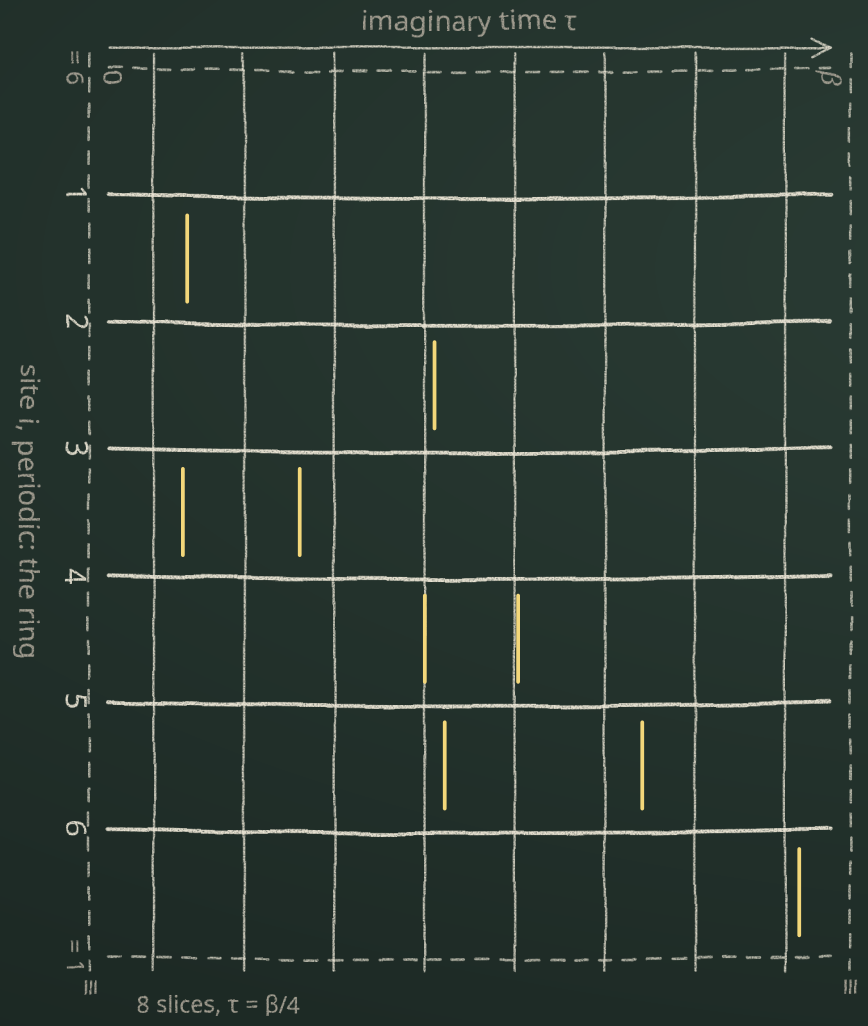
One of the 719: family 268, a proof that the spin-1 Heisenberg ring has a Haldane gap

READ	One agent refereed it from the LaTeX in a single pass and re-ran the paper's exact-arithmetic checks.
BUILD	Three agents, in parallel, each turned the paper, the referee report, an explainer and a one-page brief into a five-slide interactive explainer.
TIME	All three were back within 40 minutes. I picked one.

One partition function, two readings

$Z_n(\beta) = Tr\, e^{-\beta H_n}$, $H_n = \Sigma\, S_i S_{i+1}$ on a ring

toy: spin-½ ring, n = 6, β = 2 (keys [ ] change β, M changes the slices)



rotated: the operator now reads across bonds, one column at a time

Six worldlines, one per site. Height β. Bond events arrive as a Poisson process, rate 1 per bond per unit β: $e^{-\beta H} = e^{-\beta n/4}\, e^{\beta \Sigma Q_b}$, expanded in time-ordered singlet projectors Q_b. (Aizenman–Nachtergaele; the ticks are an illustration, the numbers below are not.)

READ ALONG TIME

Gibbs weights over 64 energy levels: $Z = \Sigma_k\, e^{-\beta E_k}$ = **657.1559026** (exact diagonalisation, uncut ring)

Cut β into 2M = 8 slices (toy lattice): $Z^{lat} = Tr[(T_A T_B)^M]$, 64 × 64 matrices = **681.135416929577**

READ ALONG SPACE

Transfer across one bond, acting on a column of 8 spins (256 × 256): $Z^{lat} = Tr[(X_e X_o)^{n/2}]$ = **681.135416929576**

Same number to 15 digits. Finer slices, M = 1, 2, 3, 4: `1136.9`, `758.17`, `700.36`, `681.14` → `657.16` (uncut).

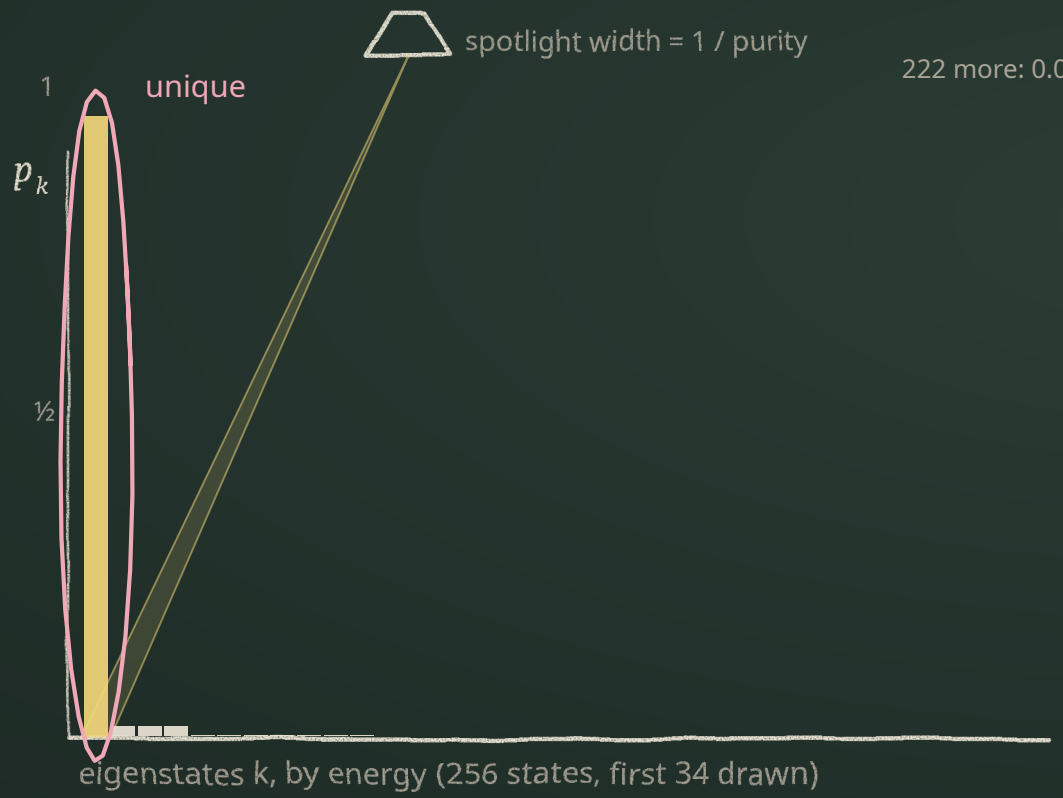
One number, two probability vectors: $p_k \propto e^{-\beta E_k}$ over levels, $q_i \propto |\lambda_i|^n$ over the transfer spectrum.

toy model: spin-½ Heisenberg ring, 6 sites, computed live · the paper's kernel is the continuous-time one, real symmetric, spin 1 · openai/math, family 268, read 7 Oct 2026

Purity as a spotlight

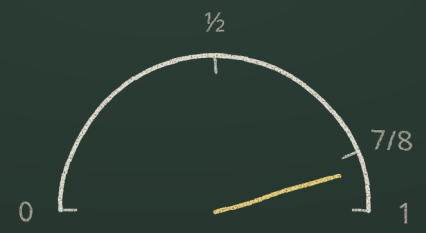
Gibbs purity $T(n,\beta) = \Sigma_k\, p_k^{\,2} = Z_n(2\beta)/Z_n(\beta)^2$

spatial purity $S(n,\beta) = \Sigma_i\, q_i^{\,2} = Z_{2n}(\beta)/Z_n(\beta)^2$ (keys [ ] for β, − = for n)



222 more: 0.000

$\beta = 8.0$



purity T

T = 0.9132

max p = 0.9553 ≥ T

defect u = 1 − T = 0.0868 < ½

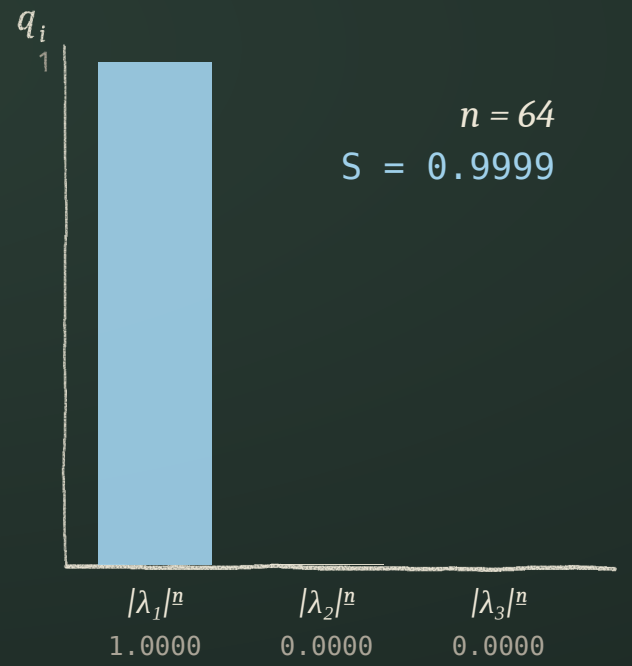
$p_1/p_0 = e^{-\beta\gamma} \le u/(1-u)$

⇒ γ ≥ 0.294 (true gap 0.523)

spatial weights, along n

classical three-state ring, β = 2 (toy)

λ = 3.116, 2.668, 0.652



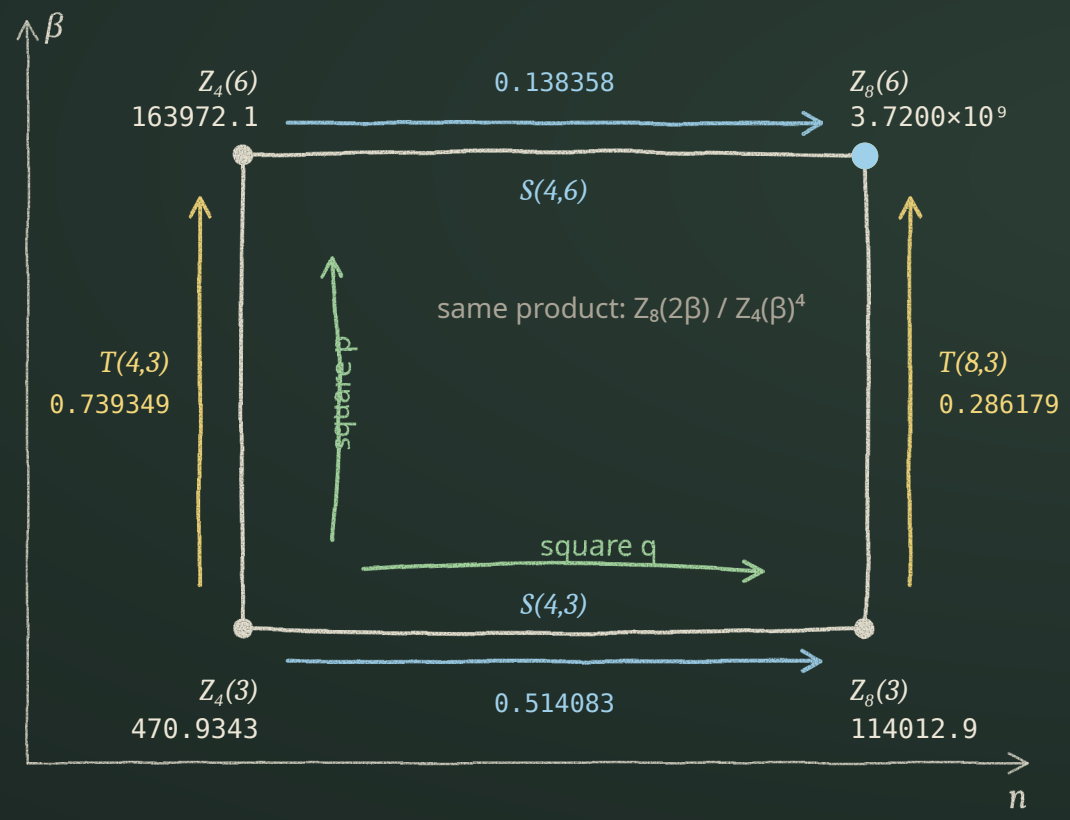
The whole problem: show $T(L,\beta)$ near 1 with β growing no faster than $\log(1/u)$.

Purity ≥ 1 − u with u < ½ forces one state of weight ≥ 1 − u: a unique ground state, and $e^{-\beta\gamma} \le u/(1-u)$.

toys: spin-½ Heisenberg ring, 8 sites, 256 states, exact diagonalisation · classical three-state ring, exact transfer matrix · openai/math, family 268, read 7 Oct 2026

The square on the blackboard

four partition functions, four purities, one identity
toy: spin-½ rings of n = 4 and $2n$ = 8 sites, β = 3 (keys [ ] change β)



THE CANCELLATION

$$S(n,2\beta)\,T(n,\beta)^2 = \frac{Z_{2n}(2\beta)}{\cancel{Z_n(2\beta)^2}} \cdot \frac{\cancel{Z_n(2\beta)^2}}{Z_n(\beta)^4} = \frac{Z_{2n}(2\beta)}{Z_n(\beta)^4}$$

$$S(n,\beta)^2\,T(2n,\beta) = \frac{\cancel{Z_{2n}(\beta)^2}}{Z_n(\beta)^4} \cdot \frac{Z_{2n}(2\beta)}{\cancel{Z_{2n}(\beta)^2}} = \frac{Z_{2n}(2\beta)}{Z_n(\beta)^4}$$

Every Z once upstairs, once downstairs. Live: left route `0.07563170461608`, right route `0.07563170461608`.

SQUARING MOVES EACH PURITY ALONG ITS OWN AXIS

Square $p_k \propto e^{-\beta E_k}$ and renormalise: the Gibbs vector at 2β. Square $q_i \propto |\lambda_i|^n$: the spatial vector at $2n$. Lemma 4.1: defect $u \to$ at most $u^2/2(1-u)^2$.

Live, Gibbs at (4, 3): u = `0.2607` → at (4, 6): `0.0148`, bound `0.0621`. Spatial, three-state ring at (8, 2): u = `0.3482` → at (16, 2): `0.1427`, bound `0.1427`.

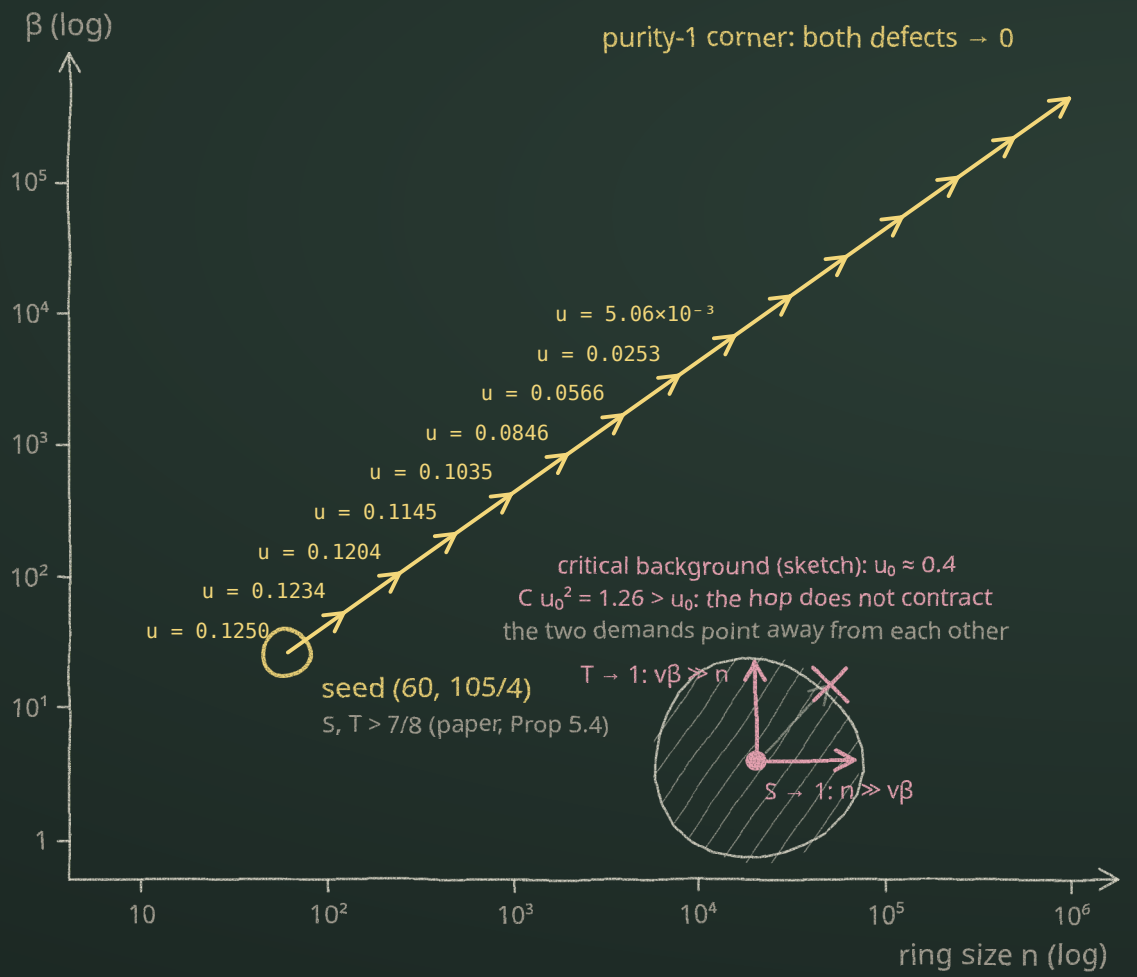
Squaring moves T along β and S along n. The identity trades one axis for the other.

toy models computed live: spin-½ rings (4 and 8 sites, exact diagonalisation), classical three-state ring · identity and Lemma 4.1 as in the paper · openai/math, family 268, read 7 Oct 2026

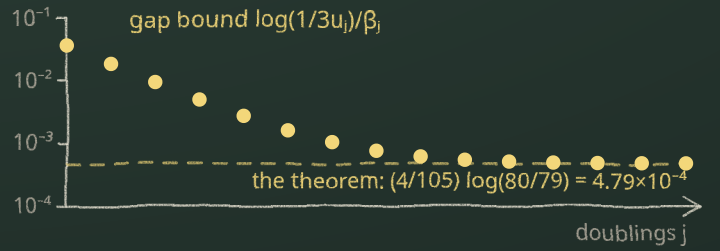
The flow

$u_{j+1} = C\,u_j^2$, $\beta_j = 2^j\beta_0$, $n_j = 2^j n_0$, gap bound $\log(1/3u_j)/\beta_j$

seed as in the paper: (60, 105/4), $u_0 = 1/8$, $C = 7.9$ (keys [ ] for u_0, − = for C)



j	n_j	β_j	u_j	$\log(1/3u_j)/\beta_j$
0	60	26.25	0.1250	0.0374
1	120	52.50	0.1234	0.0189
2	240	105	0.1204	9.70×10^{-3}
3	480	210	0.1145	5.09×10^{-3}
4	960	420	0.1035	2.78×10^{-3}
5	1920	840	0.0846	1.63×10^{-3}
6	3840	1680	0.0566	1.06×10^{-3}
7	7680	3360	0.0253	7.67×10^{-4}
8	15360	6720	5.06×10^{-3}	6.23×10^{-4}
9	30720	13440	2.02×10^{-4}	5.51×10^{-4}
10	61440	26880	3.22×10^{-7}	5.15×10^{-4}
∞			→ 0	→ $\log(1/Cu_0)/\beta_0 = 4.79\times10^{-4}$



Gapless chain: finite-size gap ≈ v/n, correlation length ≈ $v\beta$. $T \to 1$ needs $v\beta \gg n$, $S \to 1$ needs $n \gg v\beta$. Best compromise: defects 0.3 to 0.5, above the 1/8 the machine needs. Gapped: ξ and $1/\Delta$ finite, the two demands decouple.

A finite ring that looks gapped on both axes with margin 1/8 looks gapped at every larger scale. Even sizes in between: Hölder on the spatial weights.

ladder computed live from the paper's seed values (Prop 4.3, 5.4: $u_0 \le 1/8$, $C < 7.9$); critical defects 0.3 to 0.5: the review's estimate · openai/math, family 268, read 7 Oct 2026

The fine print

what is proved, what was checked, where a bug would hide

THEOREM 1.1

$H_L = \Sigma_j S_j S_{j+1}$, spin 1, periodic. Every even $L \geq 60$: unique ground state and $\gamma_L >$ **$(4/105)\log(80/79) \approx 4.8\cdot10^{-4}$**. Every even $L \geq 2304$: $\gamma_L >$ **$\log 20 / 784 \approx 3.8\cdot10^{-3}$**. Hence liminf $\gamma_L > 0$.

SCOPE

Even periodic rings and the liminf of their gaps. Cor 6.1 gives a gap inequality for subsequential limits of periodic ground states, not uniqueness or a gap for every infinite-volume ground state. True gap ≈ 0.41: the constants are a thousand times smaller.

THE SEEDS

Where the labour is. (60, 105/4) from exact Z_n, $n \leq 10$, at $\beta = 21/2$ (59,049 states, exact integers, rigorous series truncation), polynomial filters on the spectrum, a bond-dimension-26 MPS giving $Z_{120} > 1$, a 5/2-power raise of β. Margin: $S(60,105/4) > 7/8$ by $6\cdot10^{-4}$. Second seed (36, 49/4): defects 0.048 and 0.18, six doublings to (2304, 784).

VERIFICATION

No Lean. Exact-rational scalar scripts and both MPS contractions re-run and pass; tightest margins $4.7\cdot10^{-5}$ and $6\cdot10^{-4}$. Independent exact diagonalisation reproduces the tables for $n \leq 9$. Unverified: $Z_{10}(21/2)$, which the $L \geq 60$ theorem needs, and $n = 10$ to 12 at $\beta = 49/4$.

VERDICT

One reviewer, one pass: no mathematical error found. **85 to 90%** that the theorem is proved as written, conditional on the $n = 10$ to 12 integer totals. If a bug hides anywhere, it is in the seeds, not in the doubling.

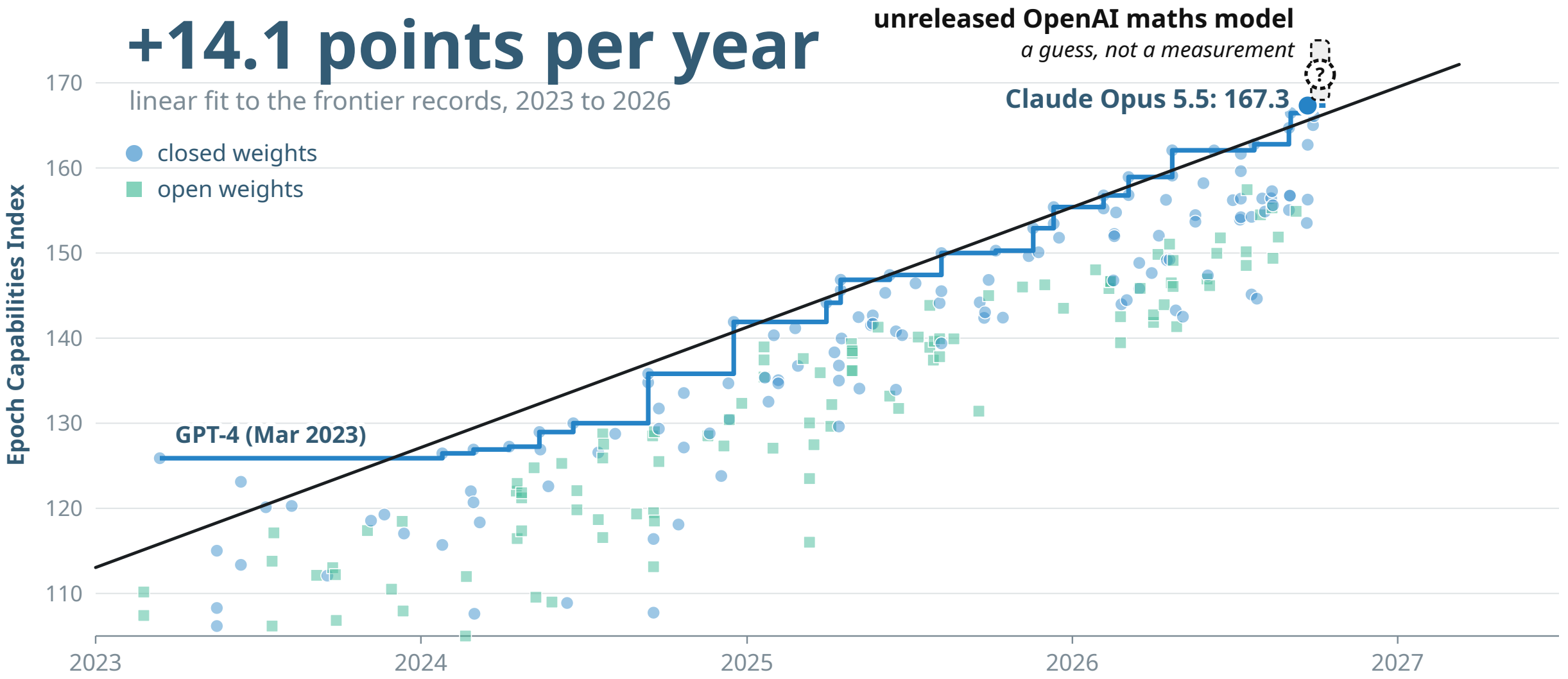
PART FIVE

The trend

Capability is a straight line

+14.1 points per year

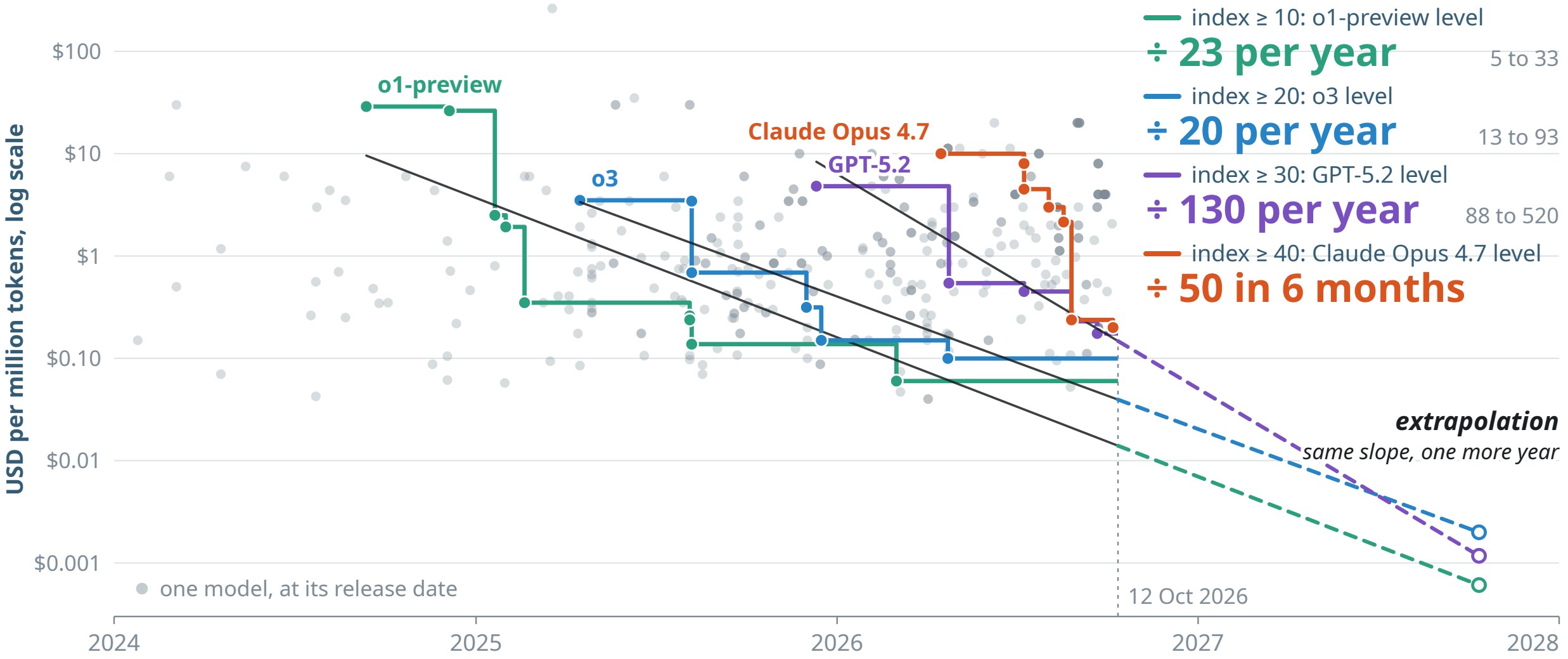
linear fit to the frontier records, 2023 to 2026



272 models, one evaluator, one scale with no ceiling.

Epoch AI Benchmarking Hub, CC-BY, snapshot 2026-10-12

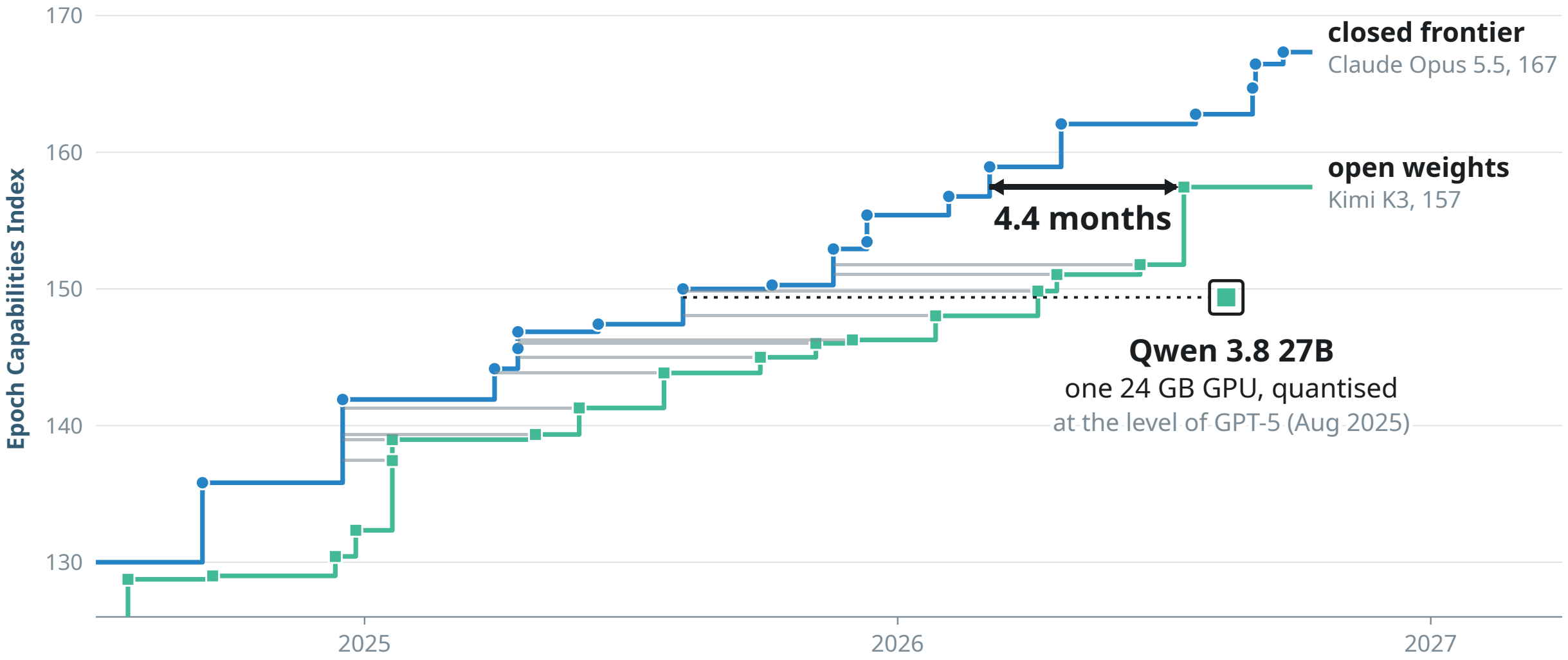
Price at fixed capability falls exponentially



Cheapest model at or above each level of the Artificial Analysis Intelligence Index.

Artificial Analysis, Index v4.3.2, snapshot 2026-10-12. Price: list, blended 3 input to 1 output. Fits and 90% ranges: T. J. Osborne

You can run the frontier locally



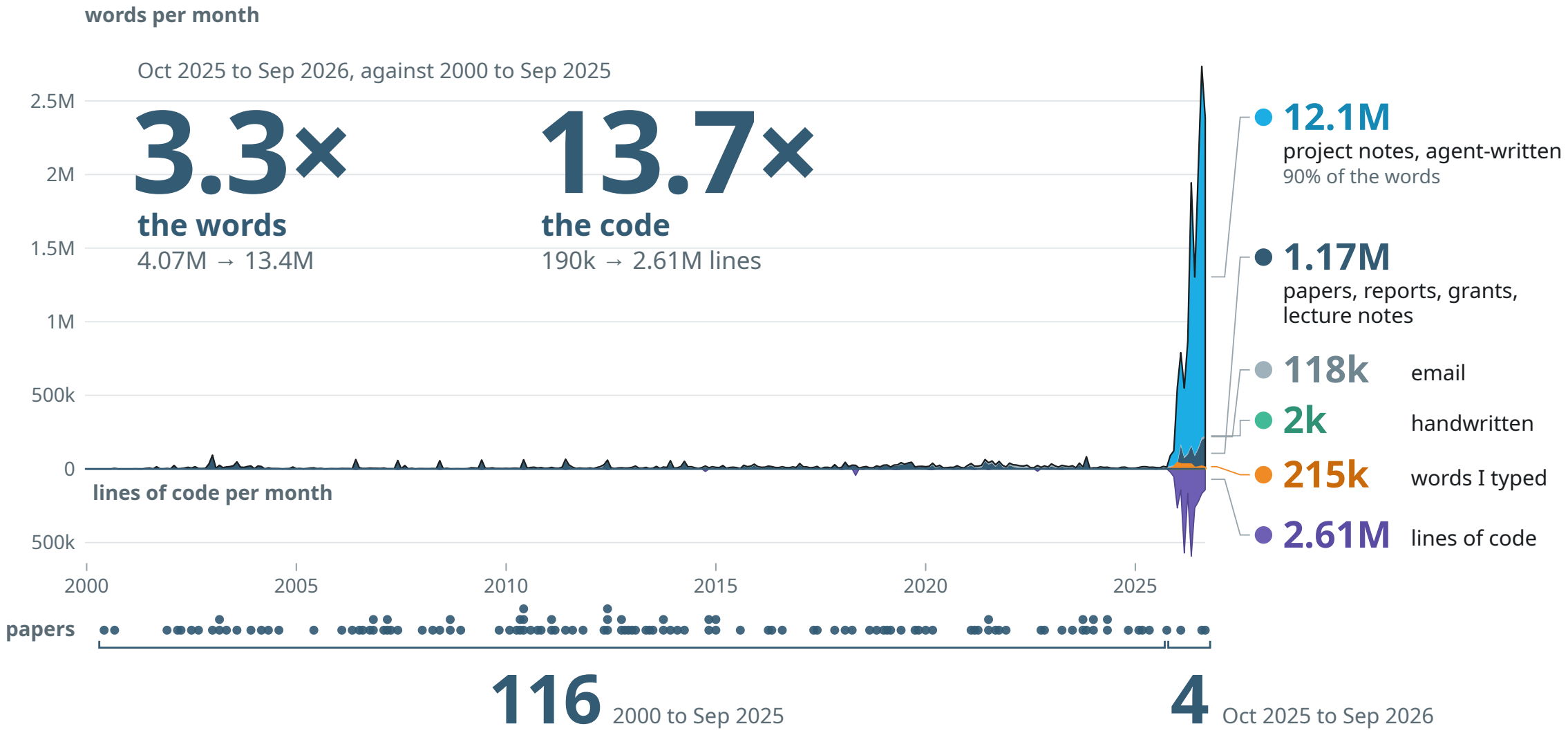
Open weights reach the closed frontier's level about 5.3 months later (median since 2025).

Epoch AI Benchmarking Hub, CC-BY, snapshot 2026-10-12

PART SIX

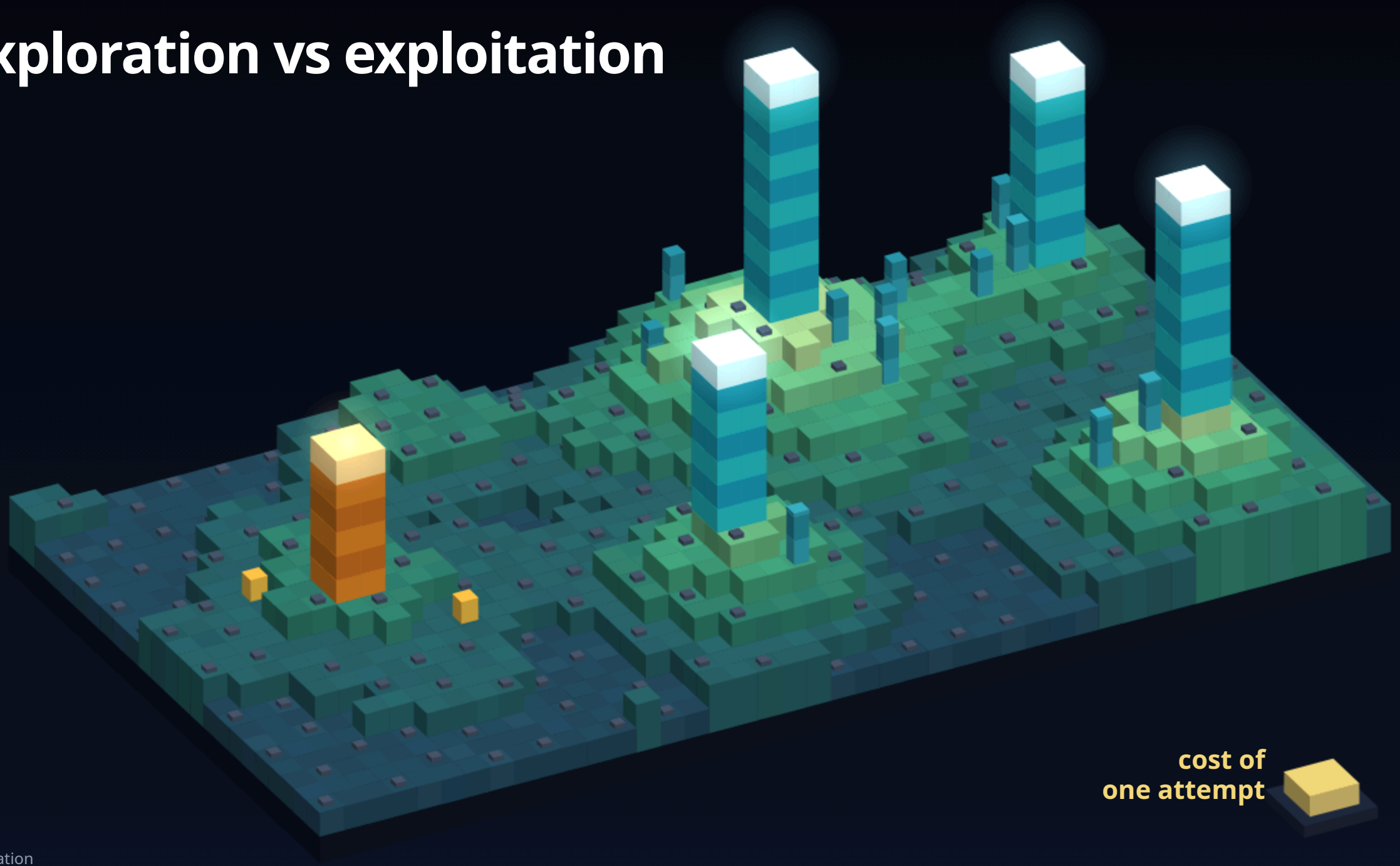
What is it for?

I measured myself



Last twelve months: 90% of the words are agent-written project notes. Duplicates, drafts, third-party text removed. Papers: INSPIRE. Method: RESULTS.md, 27 Sep 2026.

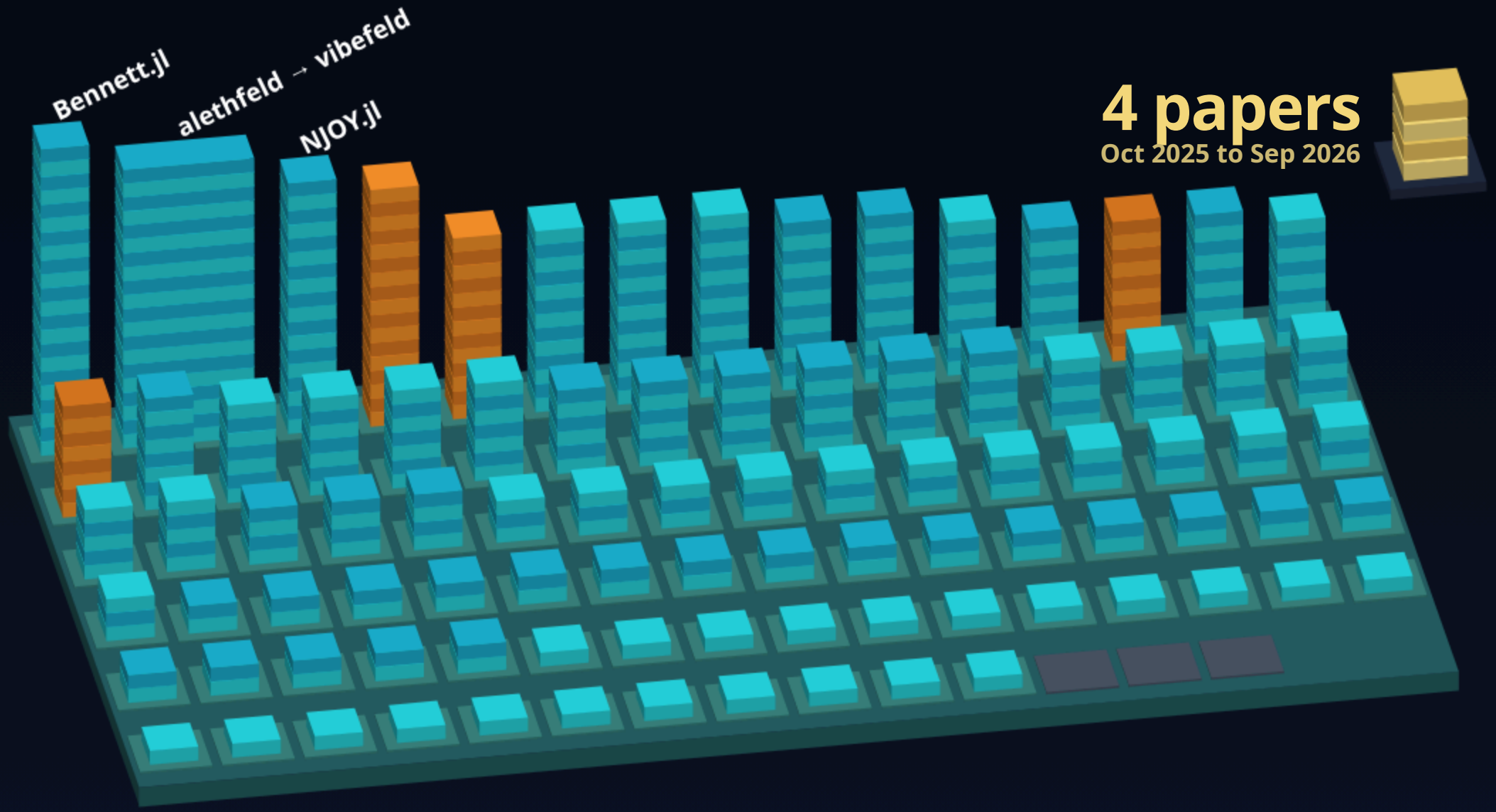
Exploration vs exploitation



illustration

94 repositories

one block: one week with commits
started 2014 to 2015



github.com/tobiasosborne, 13 Oct 2026 · grey plots: 3 forks · papers: INSPIRE

Output ≠ mission

THE WORK NOW

Effort

compute, years

Output

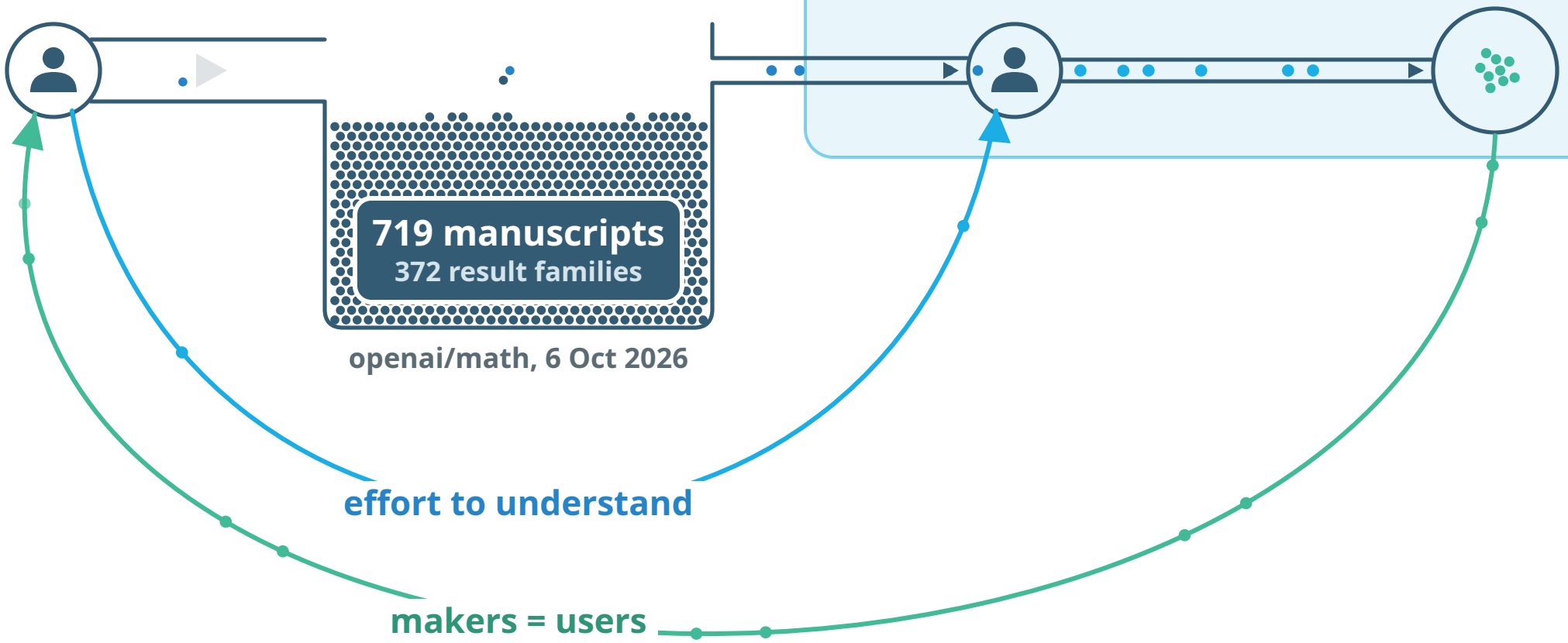
theorems, Lean certificates, problems closed

Outcome

mathematicians think, teach and ask differently

Mission

human understanding, passed on

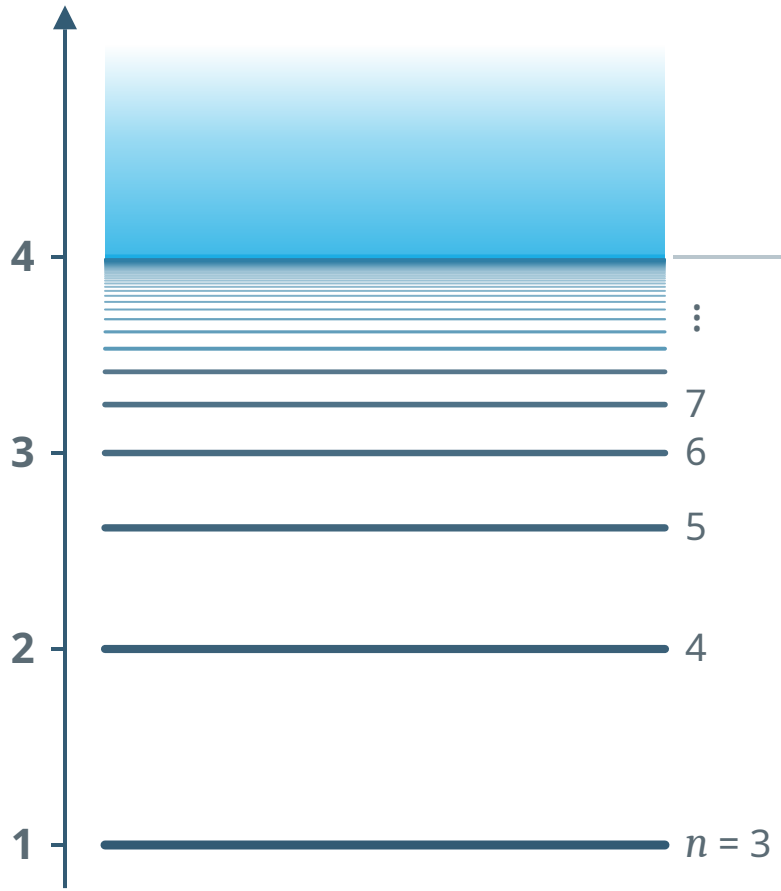


adapted from Kent Beck

After the answer

JONES, 1983

$$[M : N] \in \{4\cos^2(\pi/n) : n = 3, 4, 5, \ldots\} \cup [4, \infty]$$





1983
Temperley-Lieb, statistical mechanics



1985
Jones polynomial



1989, 1998
quantum field theory, CFT



1999
planar algebras



2003
topological quantum computation

The yield is what we build.

Jones, Invent. Math. 72 (1983) · dates and references as in Evans and Kawahigashi, arXiv:2303.04459

What works: four habits from software

01

“Ask the agent to explain things.”

“[…] you have a genius in your computer that can tell you exactly how it works.”

Theo Browne

A proof is no different.

02

Try five approaches

Being wrong is now cheap.

03

Read where a mistake matters

Statements, definitions, what others build on.

04

Make failures cheap to catch

Exact arithmetic, Lean, a second agent as referee.

“Use AI to get better.” Gergely Orosz

After Browne and Cramer (Convex Abstract panel, Sep 2026) and Orosz (LDX3 New York, 2026) · grey lines: my translation

You can outsource cognition
but you can't outsource *understanding*



The last arrow runs through us.

And we get to be here at the start.

How to train your slop cannon (Loader, Oppenheim, Osborne, 2026)

github.com/Open-Science-Ledger/how-to-train-your-slop-cannon

… when a new object comes within our reach,

the addition to our knowledge is the greater,

the more we already know;

so that it increases not as the new objects increase,

but in a much higher proportion. * * *



Colin Maclaurin
1698 to 1746

C. Maclaurin, *An Account of Sir Isaac Newton's Philosophical Discoveries* (1748), p. 392: the printed text ends here · drawing: Earl of Buchan, about 1795